

RESEARCH ARTICLE SUMMARY

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Pan-cancer spatial atlas of tertiary lymphoid structures

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INTRODUCTION: Tertiary lymphoid structures (TLSs) are ectopic immune aggregates that arise within tumors and regulate antitumor immunity. Although TLS presence has been associated with enhanced immune activity and improved outcomes in several settings, their maturation states, spatial locations relative to tumors, and context-dependent associations have not been systematically characterized at a pan-cancer scale, limiting a unified view of TLS biology and clinical utility. We asked how TLS maturation and spatial context relate to local immune programs and tumor cell states and whether these features can be quantified from routine histopathology. To address this, we combined spatial transcriptomics (ST) with deep learning applied to hematoxylin and eosin (H&E) whole-slide images (WSIs), building an atlas of TLS diversity across human cancers and a deployable framework that translates TLS spatial features into patient stratification.

RATIONALE: Whole-transcriptome spatial profiling across entire tissue sections enables unbiased identification of TLS niches, maturation trajectories, and spatial context relative to tumor regions [intratumoral (IT), peritumoral (PT), distal-tumoral (DT)], avoiding marker-limited and region of interest-constrained analyses. We uniformly processed 340 ST sections and paired histology, then extended to 3071 WSIs using an artificial intelligence (AI)-enabled framework to detect and classify TLSs at scale. We hypothesized that TLS maturation state and spatial context would be associated with distinct local immune programs and TLS proximity-linked tumor pathway patterns and that a composition-based score (capturing early, primary, or secondary TLS states and their relative proportions) would outperform binary “TLS present or absent” or dominant-state heuristics for prognosis. Cross-cohort analyses, orthogonal validations (including ultrahigh-plex imaging and single-cell spatial profiling), and sensitivity tests were used to assess robustness and generalizability.

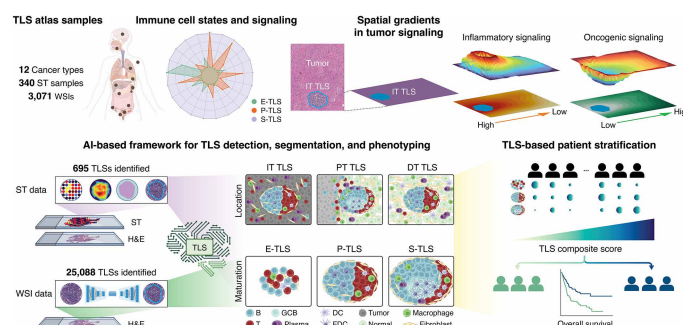
RESULTS: Across 12 cancer types, we profiled TLSs including TLSs resolved by ST and identified from WSIs. TLSs varied substantially in prevalence, maturation, and spatial distribution across cancer types. TLSs spanned a maturation continuum [early (E-TLSs), primary follicle-like (P-TLSs), and secondary follicle-like (S-TLSs)]

and occupied distinct spatial niches relative to tumors. Maturation was associated with coordinated immune organization: B and T cell zoning and activation, follicular dendritic cell networks, cytokine and chemokine signaling, and interferon (IFN)-related responses. Spatial analyses around intratumoral TLSs (IT-TLSs) revealed distance-dependent tumor pathway gradients: Immune activation and antigen-presentation programs (e.g., IFN- α and IFN- γ responses and MHC class II signatures) were enriched in tumor regions closest to TLSs and attenuated with distance, whereas proliferative and invasive programs (e.g., MYC targets, G2/M cell-cycle programs, epithelial-mesenchymal transition, and related oncogenic pathways) were relatively higher in TLS-distal tumor regions, highlighting spatial coupling between TLS niches and tumor programs. These findings were supported by single-cell-resolved spatial multimodal profiling. To scale beyond ST, we trained an AI framework to detect TLSs and predict maturation states directly from H&E WSIs and evaluated it across additional The Cancer Genome Atlas and independent therapy cohorts. We distilled these spatial features of TLSs into a maturation-aware composite score capturing within-tumor TLS state composition that robustly stratifies patients by survival and treatment response across cancers and treatment contexts.

CONCLUSION: TLSs are spatially organized, heterogeneous immune hubs whose maturation and spatial context are linked to local immune organization, tumor pathway gradients, and patient outcomes. By uniting whole-section ST with AI on standard WSIs, we provide both a pan-cancer atlas of TLS diversity and a practical route to scalable, histology-deployable TLS profiling. The composition-based TLS composite score is positioned for prospective evaluation as a stratifier or endpoint in immuno-oncology trials and may help guide strategies aimed at promoting beneficial TLS states. Future work should incorporate longitudinal sampling under therapy, improved subclass or isotype-resolved antibody profiling as platforms mature, and functional perturbation to test causal mechanisms and refine TLS-informed clinical decision-making. □

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Pan-cancer TLS atlas and AI-enabled patient stratification. We built a pan-cancer TLS atlas and developed AI-enabled frameworks to detect and phenotype TLSs. Across 12 cancers, we mapped TLS location, maturation states, and tumor-signaling gradients. A maturation-aware composite score stratifies patients across clinical contexts. GCB, germinal center B cell; DC, dendritic cell; FDC, follicular dendritic cell.



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Pan-cancer spatial atlas of tertiary lymphoid structures

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Tertiary lymphoid structures (TLSs) are critical regulators of antitumor immunity, yet their spatial organization, maturation, and clinical relevance remain incompletely defined across cancers. We analyzed spatial transcriptomics spanning 12 cancer types to construct a pan-cancer TLS atlas and characterized TLS spatial architecture and maturation states. TLS maturation was accompanied by coordinated remodeling of distinct niche cell populations and distance-dependent gradients in tumor programs, orthogonally supported by ultrahigh-plex single-cell spatial profiling. To enable scalable TLS profiling, we trained an artificial intelligence framework that predicts TLS maturation states directly from hematoxylin and eosin–stained images and evaluated it across TCGA and independent therapy cohorts. We further derived a maturation-aware composite score capturing intratumoral TLS state composition, which robustly stratifies patients across cancer and treatment contexts, outperforming conventional TLS metrics.

Effective antitumor immunity in solid tumors depends on orchestrated local immune microenvironments that support lymphocyte activation. Tertiary lymphoid structures (TLSs) are ectopic lymphoid aggregates that arise in nonlymphoid tissues under chronic inflammatory conditions, including cancer (1). TLSs frequently develop within the tumor microenvironment (TME) and have been observed across a broad range of human solid tumors, where they contribute to lymphocyte activation, B cell immunity, and regulation of antitumor immune responses (2, 3). Intratumoral TLSs support B cell maturation and antibody production (3) and are associated with increased tumor-infiltrating lymphocytes, particularly effector memory CD8⁺ T cells (4). The presence of TLSs has been linked to favorable responses to immune checkpoint blockade (ICB) and prolonged survival across multiple cancer types

(4–11), fueling interest in TLSs as predictive biomarkers, prognostic indicators, and potential therapeutic targets (11–13).

Resembling secondary lymphoid organs, TLSs typically contain segregated B cell follicles and T cell zones, antigen-presenting cells, and specialized stromal components, including fibroblasts (14) and high endothelial venules (HEVs), which facilitate immune cell recruitment and spatial organization (3). Mature TLSs containing germinal centers (GCs) are generally associated with enhanced antitumor immunity and improved ICB efficacy (10, 15). However, accumulating evidence indicates that TLSs are highly heterogeneous in their spatial distribution, cellular composition, organization, and maturation state, both within and across anatomical sites and tumor types (16). TLSs can range from early lymphocyte aggregates to fully developed secondary follicle-like structures with GCs (16, 17), and their formation and maturation appear to be influenced by tumor antigenicity and local TME dynamics (2, 16). Tumor-intrinsic factors, including metabolic reprogramming, have also been implicated in constraining TLS maturation in specific cancer contexts (18).

Despite these advances, a systematic characterization of TLS phenotypic and functional heterogeneity in large tissue sections across cancer types remains lacking. Moreover, how TLS profiles relate to tumor molecular features and clinical outcomes across large cohorts and treatment contexts remains incompletely defined. Addressing these gaps is essential for understanding context-dependent mechanisms of TLS formation and function, laying a foundation for TLS-based biomarker development and informing therapeutic strategies aimed at modulating TLSs. Spatial profiling technologies provide new opportunities to study TLSs. For example, spatial transcriptomics (ST) now enables transcriptome-wide profiling across entire tissue sections, typically paired with high-resolution hematoxylin and eosin (H&E) images (19, 20), allowing detailed examination of TLSs within large tissue samples. However, scalable, integrated computational frameworks for automated TLS detection and phenotyping from spatial omics and histology data are currently lacking. Additionally, current approaches for TLS-based patient stratification are suboptimal. Most studies categorize patients based on the presence or absence of TLSs (21–23) or use broad groupings, such as mature versus immature TLSs (21, 24), which may not be sufficiently robust because they do not capture intratumoral heterogeneity in TLS states. There is a need to establish robust, scalable metrics to enable more precise patient stratification across diverse clinical settings and enhance the utility and impact of TLS-based biomarkers.

In this study, we developed a robust computational framework to automatically identify, segment, and phenotype TLSs across cancer types and applied it to a pan-cancer analysis. By integrating transcriptomic, spatial, histopathological, and clinical data, we systematically characterized TLS abundance, spatial distribution, size, maturation states, and transcriptomic programs in 340 ST samples across 12 cancer types and examined their interactions with tumor cells and the surrounding TME. Building on insights from paired spatial omics and histology, we trained a deep learning model to classify TLSs using H&E-stained whole-slide images (WSIs) alone and applied it to classify 25,088 TLSs from 3071 WSIs across 10 independent cohorts. Lastly, we

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developed a data-driven, unsupervised TLS-based patient stratification framework that outperformed existing approaches in prognostic evaluation. Together, this work provides a comprehensive landscape of TLS heterogeneity across cancers and establishes spatially defined TLS features and artificial intelligence (AI)-driven TLS classification as scalable tools for precision immuno-oncology.

ST reveals the landscape of TLSs across tissues

To systematically characterize the spatial and molecular heterogeneity of TLSs in cancer, we analyzed ST data from 340 tumor samples collected from 255 patients across 12 cancer types, incorporating both publicly available and in-house datasets (Fig. 1A and tables S1 and S2). All datasets were processed uniformly using our established computational framework (Fig. 1B; see materials and methods). TLS-enriched regions were identified based on high TLS signature scores derived from a curated TLS gene set (25) (table S3), followed by unsupervised clustering, segmentation, and filtering. Each segmented TLS was automatically assigned a unique identifier, yielding 695 TLSs for downstream

analyses, including assessment of maturation state, spatial location, and transcriptomic profiling (Fig. 1B). The proportion of samples containing TLSs varied substantially across cancer types (Fig. 1C). TLSs were most prevalent in lung adenocarcinoma (LUAD), kidney renal clear cell carcinoma (KIRC), stomach adenocarcinoma (STAD), breast cancer (BRCA), liver hepatocellular carcinoma (LIHC), and bladder carcinoma (BLCA), whereas pancreatic adenocarcinoma (PAAD), oral squamous cell carcinoma (OSCC), ovarian carcinoma (OVCA), and glioblastoma multiforme (GBM) exhibited few or no detectable TLSs. TLS size and frequency within individual tumors also differed markedly across cancer types (Fig. 1D and table S4). LUAD samples exhibited the highest number of TLSs per sample, whereas no TLSs were detected in the 36 GBM samples analyzed, despite prior evidence that TLSs can form in gliomas (26). The absence may reflect limited ST dataset availability and sampling depth for GBM. We further classified TLSs into three spatial categories—intratumoral (IT), peritumoral (PT), and distal-tumoral (DT)—based on their spatial proximity to tumor regions (see materials and methods). TLSs were

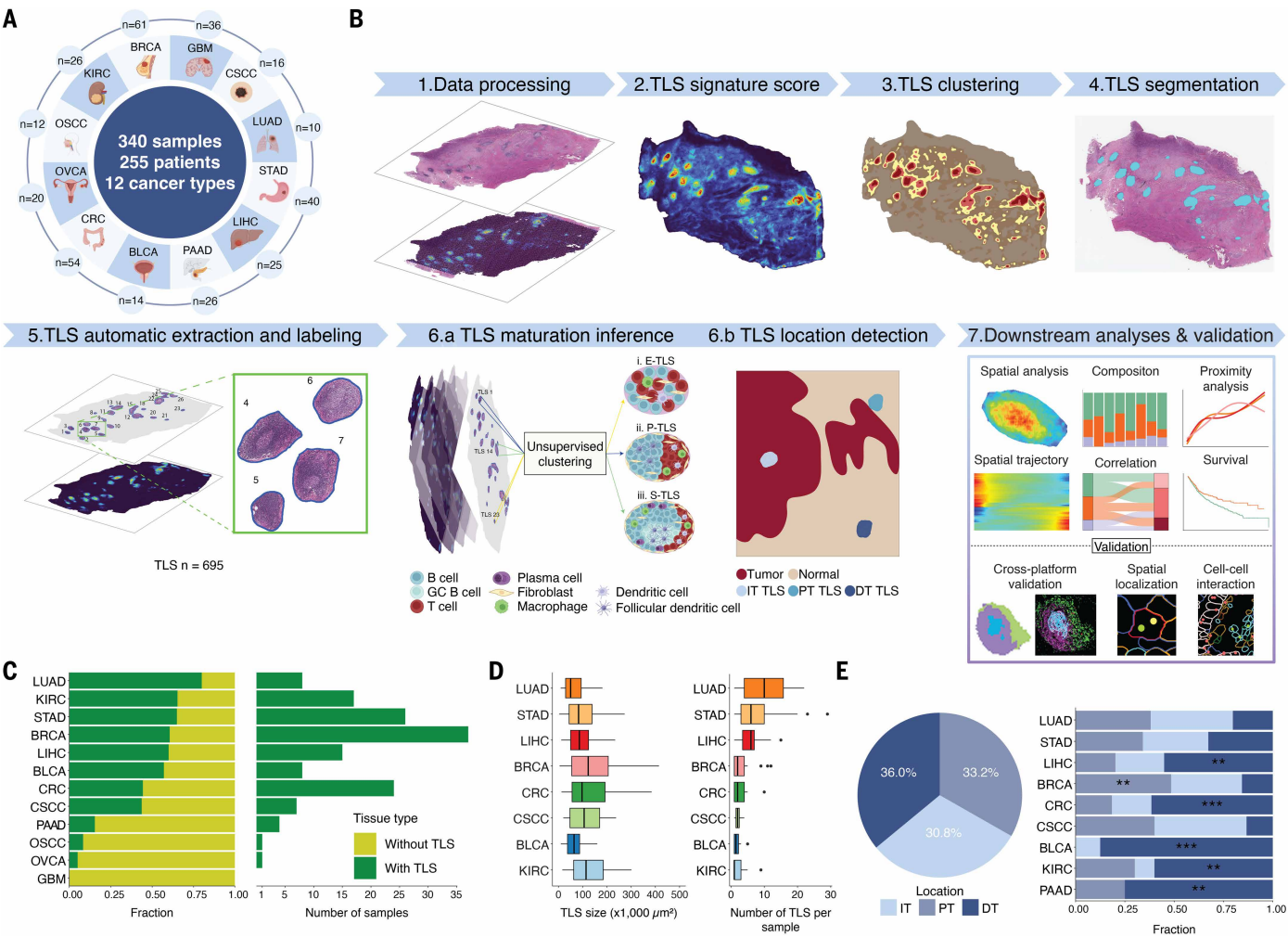


Fig. 1. Study design and workflow for TLS detection and analysis. (A) Schematic of the study design. TCGA cancer type abbreviations are used. (B) Overview of the analysis workflow (see materials and methods). Data are input into iStar (25), and a TLS signature score is calculated based on our previously reported TLS signature (25). Unsupervised clustering is performed to extract TLS regions, followed by TLS segmentation. Gene expression data are extracted simultaneously for each TLS and TLS IDs are automatically assigned, followed by maturation inference, spatial location identification, and downstream analyses. IT, TLS within tumor tissue. PT, TLS within two spots (200 μ m) of tumor. DT, TLS greater than two spots from tumor. (C) (Left) Fractions of samples with or without TLS. (Right) Number of TLS⁺ samples across 12 cancer types. (D) (Left) Average TLS size. (Right) Number of TLS per sample by cancer type. (E) (Left) Proportion of TLSs by cancer type location. (Right) Relative fractions of TLS locations across cancer types. *** $P < 0.001$; ** $P < 0.01$; * $P < 0.05$; P values were calculated using Pearson's χ^2 test. In (D, E) right panels, cancer types with > 5 TLS⁺ samples were included.

distributed relatively evenly across the three categories, with 214 IT, 231 PT, and 250 DT TLSs identified, although their relative proportions varied by cancer type (Fig. 1E and table S4). LUAD and CESC showed higher fractions of IT TLSs, whereas BRCA was enriched for PT TLSs, and LIHC, colorectal cancer (CRC), BLCA, KIRC, and PAAD exhibited higher fractions of DT TLSs. Together, these analyses reveal pronounced heterogeneity in TLS prevalence, size, and spatial organization across cancers, underscoring the diverse architectural patterns of TLSs within the TME.

TLS maturation states differ across cancer types

To characterize TLS maturation, we classified TLSs into three maturation states—early TLS (E-TLS), primary follicle-like TLS (P-TLS), and secondary follicle-like TLS (S-TLS)—using unsupervised clustering based on canonical markers of follicular dendritic cells (FDCs) and germinal center (GC) B cells (Fig. 2A and table S4; see materials and methods), consistent with established models of TLS development (1, 16). This analysis identified 379 E-TLSs (54.5%), 215 P-TLSs (31%), and 101 S-TLSs (14.5%). E-TLSs represent early lymphoid aggregates lacking organized follicular structure, P-TLSs exhibit segregated B- and T-cell zones with FDC presence but without a fully developed GC, and S-TLSs are characterized by well-organized B- and T-cell zones with established FDC networks and a localized GC (Fig. 2B and fig. S1). Consistent with progressive maturation, TLS signature scores increased stepwise from E-TLS to S-TLS, accompanied by elevated expression of markers for T cells, B cells, FDCs (e.g., *CR2*, *FCER2*) and GC B cells (GCBs; e.g., *BCL6*, *MKI67*) (Fig. 2C). To validate these assignments, we generated paired ST (Visium, 10x Genomics) and spatial proteomics (SP, Lunaphore COMET or Akoya IO60) on the same or adjacent tissue sections (table S5; see materials and methods). ST-defined TLS architecture closely concurred with protein-level staining of key immune and stromal components (Fig. 2D and fig. S2), including B cell follicles, T-cell zones, FDC networks (fig. S2A), and GC features as well as additional TLS components, such as T follicular helper (Tfh) cells and fibroblasts (fig. S2B). Notably, in P-TLSs and S-TLSs, we consistently observed a CD21⁺ reticular or mesh-like pattern localized to the B cell follicular region, characteristic of FDC networks (Fig. 2D and fig. S2A). These features were reproducibly detected across spatial omics platforms and independently reviewed by board-certified pathologists, providing orthogonal histologic support for the robustness and biological validity of our TLS maturation assignments.

We next examined the spatial distribution of TLS maturation states across IT, PT, and DT regions (Fig. 2E). P-TLSs were preferentially enriched in DT regions, whereas E-TLSs were more frequently observed within IT regions (Fig. 2E). TLS size increased progressively with maturation state, with S-TLSs significantly larger than P-TLSs and E-TLSs (Fig. 2F and fig. S2C). Analysis across cancer types revealed marked heterogeneity in TLS maturation states (Fig. 2, G and H). S-TLSs were enriched in STAD, P-TLSs were enriched in LIHC and CRC, and E-TLSs were enriched in BRCA and OVCA (Fig. 2H). Integrating maturation state with spatial location further delineated nine TLS subtypes, revealing additional layers of heterogeneity across tumor types. For example, DT P-TLSs predominated in LIHC, CRC, and BLCA; PT E-TLSs predominated in OVCA, BRCA, and LUAD; and DT E-TLSs predominated in PAAD and KIRC (Fig. 2I). Together, these results demonstrate that TLS maturation and spatial organization vary substantially across cancers, highlighting the context-dependent nature of TLS development.

Transcriptional programs associated with TLS maturation

We then investigated transcriptional differences across TLS maturation states by comparing the expression of markers and gene signatures across key cellular components of TLSs as well as cytokines and chemokines and related signaling programs in E-TLSs, P-TLSs, and S-TLSs (Fig. 3A and fig. S3). These analyses revealed coordinated

and stage-dependent remodeling of the immune and stromal components of TLSs as they progress toward functional maturation.

B cell programs evolved progressively with TLS maturation, reflecting enhanced B cell recruitment, organization, activation, and differentiation. P-TLSs exhibited elevated expression of canonical B cell markers and genes associated with B cell activation and early differentiation (e.g., *BANK1*, *CD40*) along with plasma cell (PC)-related markers (e.g., *IGHA1*, *IGHG1*, *MZB1*, *JCHAIN*). By contrast, S-TLSs were characterized by enrichment of GCB cell, memory B cell, and PC differentiation programs (e.g., *PRDM1*, *XBPI*, *SDCI*, *CD38*), consistent with advanced B cell maturation, class switching, and antibody production (27) (Fig. 3A). The lower abundance of PC-associated transcripts within S-TLS cores likely reflects dissemination of terminally differentiated antibody-secreting cells (ASCs) from mature TLSs.

To directly assess ASC abundance and isotype distribution, we performed multiplex immunofluorescence in clear cell renal cell carcinoma (ccRCC), a nonmucosal context chosen to minimize tissue-of-origin immunoglobulin A (IgA) skewing (table S6; see materials and methods). The panel combined PNAd, CD20, CD23, and BCL6 to define TLS architecture and maturation together with MUM1 (IRF4), IgA, and IgG to assess ASC identity and isotype distribution (Fig. 3B). In E-TLSs, ASCs were sparse, largely confined within TLS boundaries, and predominantly IgG⁺ (Fig. 3B, top). By contrast, P-TLSs and S-TLSs exhibited increased ASC density, with both IgG⁺ and IgA⁺ populations frequently intermingled within follicular regions and disseminated into peri-TLS and adjacent tumor areas (Fig. 3B, middle and bottom). These findings support a functional link between TLS maturation and antibody output.

In parallel, TLS maturation was accompanied by increased lymphoid recruitment and niche organization signals, including CCR7-axis chemokines (e.g., *CCL19*) that support lymphocyte trafficking and positioning; *CXCL13*, associated with B cell recruitment and follicular organization; and immune-activating cues (e.g., *CXCL10*, *IL7*, *IL33*) that were most elevated in S-TLSs (Fig. 3A and fig. S3A) (11, 28, 29). S-TLSs also showed increased expression of proliferation-associated genes (e.g., *MKI67*, *BIRC5*, *STMN1*), consistent with active cellular expansion within mature TLS niches.

Notably, regulatory B cell (Breg)-associated genes (*IL10*, *TGFB1*, *CD24*, *CD5*, *CD1D*) peaked in S-TLSs, indicating engagement of immunoregulatory programs alongside adaptive immune activation (27). To define the cellular sources of immunoregulatory cytokines within TLSs, we analyzed in-house Xenium Prime 5K ST data from LUAD samples, which provides single-cell and subcellular resolution for in situ mapping of cytokine transcripts within defined B cell populations (see materials and methods). We identified a subset of B cells coexpressing *IL10* and *CD24* and/or *CD1D*, referred to here as Breg-like cells (Fig. 3C and fig. S3B). These cells were spatially enriched within S-TLSs, providing direct spatial evidence that IL10-expressing B cells localize to mature TLSs. Notably, *IL10* and *TGFB1* expression in S-TLSs may not be restricted to B cells, as these cytokines can also be produced by myeloid or stromal cells within or adjacent to TLSs.

As TLSs matured, T cell programs diversified into distinct functional states. E-TLSs showed limited T cell activation, whereas mature TLSs, particularly S-TLSs, were enriched for memory, effector, and cytotoxic T cell signatures (Fig. 3A and fig. S3A). S-TLSs also showed enrichment of follicular helper T cells (Tfh), supported by coexpression of CD4, ICOS, and PD-1 proteins detected by high-plex imaging (IO60 panel; fig. S2B), together with increased expression of canonical Tfh markers (Fig. 3A) and a Tfh gene signature (30) (fig. S3C). These states were accompanied by elevated recruitment cues (Fig. 3A; fig. S3C) and increased interferon activity, reflected by elevated interferon-stimulated genes and key regulators of interferon signaling, such as *STAT1* (31) (fig. S3D). In parallel, S-TLSs showed increased expression of regulatory T cell markers and immune checkpoint and exhaustion programs (e.g., *FOXP3*, *IL2RA*, *PDCD1*, *CTLA4*, *TIGIT*, *LAG3*, *HAVCR2*), indicating that

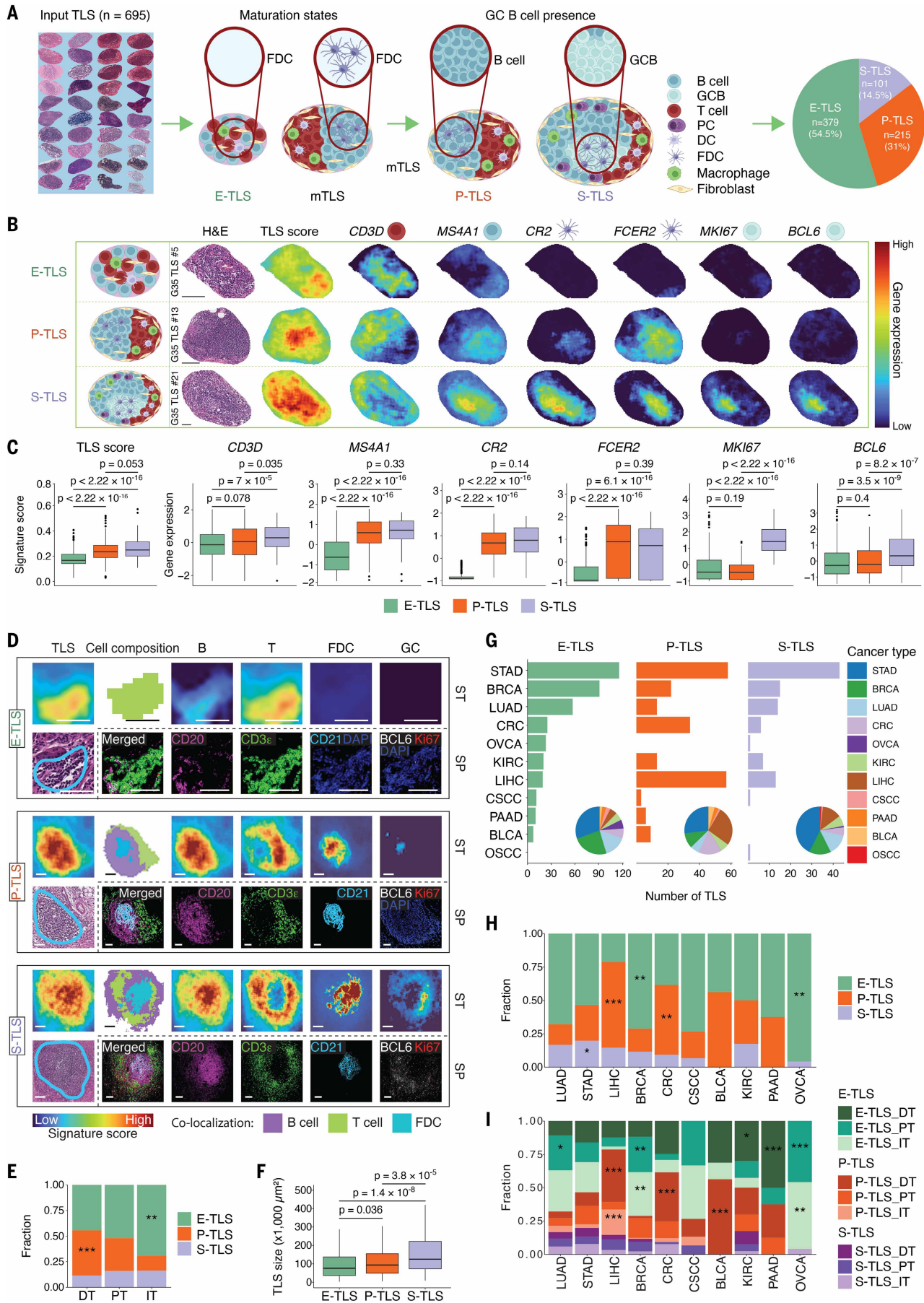


Fig. 2. Systematic classification and spatial profiling of TLS maturation states. (A) Workflow for TLS maturation classification. TLSs are first divided into early (E-TLS) and mature TLSs (mTLS) based on the presence of follicular dendritic cells (FDCs). FDC⁺ TLSs are further categorized as P-TLS or secondary follicle-like TLS (S-TLS) depending on germinal center (GC) presence. Pie chart displays the proportions and counts of each maturation state. (B) Representative examples of E-TLS, P-TLS, and S-TLS, each with an H&E image and corresponding heatmaps showing differential expression of TLS score and six marker genes used for maturation classification. Scale bar, 100 μ m. (C) Differential expression of TLS score and six marker genes across three TLS maturation states. All TLSs are included. *P*-values from Wilcoxon rank-sum tests are shown. (D) Cross-platform validation of TLS maturation state assignments using ST (Visium ST) and SP (COMET seqIF). Representative E-TLS, P-TLS, and S-TLS examples are shown from adjacent tissue sections of the same tumors. Top row (Visium ST), left to right: H&E image, TLS signature score, cell colocalization map, and cell-type signature scores for B cells, T cells, FDCs, and germinal center B cells. Bottom row (COMET), left to right: merged immunofluorescence composite and single-channel images for CD20 (B cells), CD3 ϵ (T cells), CD21 (FDCs), and BCL6/Ki67 (GC), highlighting a CD21⁺ reticular pattern in mTLSs. Scale bar, 100 μ m. (E, H, and I) Relative fractions of TLS maturation states across spatial locations (E), cancer types (H), and combined characterization of TLSs by maturation state and spatial location (I). (F) Boxplots comparing the TLS size across maturation states. *P* values from Wilcoxon rank sum tests are shown. (G) TLS counts by maturation state across cancer types. Bar plots and pie charts illustrate the distributions. Cancer types with >1 TLS were included. ****P* < 0.001; ***P* < 0.01; **P* < 0.05; *P* values were calculated by Pearson's χ^2 test.

mature TLSs couple immune activation with regulatory restraint. Up-regulation of stress-response genes (e.g., *HSPA1A/B*, *HSPA6/8*, *HSPD1*, *BAG3*, *DNAJB1/4*) in S-TLSs further suggests metabolic adaptation accompanying sustained immune activation (21, 28, 31, 32) (fig. S3D).

Beyond lymphocytes, TLS maturation was accompanied by coordinated remodeling of antigen-presenting, stromal, and vascular compartments. As TLSs progressed from E-TLSs to S-TLSs, we observed increased expression of DC-associated markers (e.g., *CD1C*, *CD86*, *LAMP3*), DC-recruitment cues (*CCL19*, *CCL21*, *CCR7*), and cytokines (*IL12A*, *IL27*) together with enrichment of macrophage activation programs and macrophage-attracting chemokines (*CCL2*, *CCL4*, *CXCL16*) and cytokines (*IL10*, *IL6*) and along with stronger antigen-presentation signals (Fig. 3A and fig. S3D) (29, 33–36). In parallel, vascular and stromal remodeling intensified with TLS maturation, reflected by increased expression of vascular and lymphatic endothelial markers (*LYVE1*, *CD34*), angiogenic factors (*VEGFA*, *KDR*, *TXNDC5*), and associated signaling molecules (e.g., *CXCL12*, *CXCR4*, *IL6*, *IL11*, *OSM*). Consistently, HEVs were observed in histology and spatial proteomics images. Together, these coordinated changes support the view that mature TLSs couple antigen presentation and innate immunity with stromal and vascular support that collectively sustain highly organized functional immune niches in the TME (14).

TLS proximity associates with spatial gradients of tumor signaling

To investigate spatial associations between tumor programs and TLS characteristics, we correlated tumor-pathway activity in tumor-enriched spots with sample-level TLS features, including E-, P-, and S-TLS proportions and TLS location metrics, as well as location-stratified maturation measures (see materials and methods). These analyses were designed to characterize spatial associations between TLS features and tumor signaling rather than infer causal effects of TLSs on tumor programs. Across cancers, TLS maturation state and spatial location were associated with distinct tumor pathway profiles (fig. S4A). E-TLSs generally showed positive correlations with proliferative and inflammatory pathways and negative correlations with metabolic and stress-adaptive programs in tumor spots. By contrast, more mature TLSs, particularly PT P-TLSs and S-TLSs, exhibited attenuated associations with proliferative and inflammatory tumor programs relative to E-TLSs and instead showed positive associations with immune-related pathways, including complement activation and interferon (IFN)- γ and IFN- α responses. In addition, IT S-TLSs showed negative associations with tumor-related programs, such as proliferation, epithelial-mesenchymal transition (EMT), angiogenesis, and MYC signaling. P-TLSs displayed location-dependent patterns, with PT P-TLSs more closely resembling S-TLS-associated profiles. Together, these results indicate that TLS maturation and spatial context are linked to distinct tumor signaling environments, highlighting maturation- and location-dependent heterogeneity in TLS-associated tumor-microenvironment interactions.

We then examined how TLS proximity shapes tumor signaling by modeling pathway activity as a function of spatial distance from IT TLSs (Fig. 4A). Two dominant spatial gradients emerged across cancers (Fig. 4, B to D). Immune-related pathways, including IFN- α and IFN- γ responses, major histocompatibility complex class II antigen presentation, and inflammatory signaling, were highest in tumor regions proximal to TLSs and progressively decreased with distance (“high-to-low” gradients). By contrast, tumor-intrinsic programs associated with proliferation and invasion, such as MYC signaling, G2/M checkpoint, EMT, and KRAS signaling, were lowest near TLSs and increased with distance (“low-to-high” gradients) (Fig. 4C). These spatial patterns were conserved across multiple cancer types (Fig. 4, C and D). When applying the same distance-based framework to TLSs located in histologically normal tissues, immune-related pathways retained similar proximity-dependent gradients, whereas tumor-intrinsic proliferative and invasive gradients were absent (fig. S4B). This contrast indicates that TLS-associated immune activation represents a conserved spatial feature of TLS niches, whereas distance-dependent tumor-associated pathway gradients are specific to malignant tissue contexts.

To explore potential signaling interactions at the TLS-tumor interface, we analyzed ligand-receptor pairs enriched within IT TLSs’ boundaries by comparing gene expression within IT TLS cores to their immediate 50- μ m margins (see materials and methods; fig. S4C). Several immune signaling genes, including *TNFSF14*, *CXCR3*, and *CXCR6*, were preferentially expressed within TLS cores, whereas their cognate receptor or ligands (*TNFRSF14*, *CXCL10/11*, and *CXCL16*) exhibited pronounced distance-dependent “high-to-low” gradients in surrounding tumor regions, peaking at TLS-proximal tumor spots and progressively decreasing with increasing distance across multiple cancer types (Fig. 4E). These patterns are consistent with the presence of short-range cytokine and chemokine signaling fields associated with TLSs.

We further validated these spatial relationships at single-cell resolution using paired Visium and Xenium Prime 5K datasets from our in-house LUAD cohort (Fig. 4F and fig. S4D; see materials and methods) (37). TLS cores were enriched for *TNFSF14*, *CXCR3*, and *CXCR6* transcripts, whereas *TNFRSF14*, *CXCL10/11*, and *CXCL16* localized predominantly to tumor or stromal cells immediately adjacent to TLSs (Fig. 4G). This complementary spatial organization, mapped at single-cell and subcellular resolution, supports localized inflammatory signaling patterns centered on TLSs and engagement of cognate signaling partners within TLS-proximal tumor regions. Together, these results indicate that TLSs are embedded within short-range cytokine and chemokine signaling fields characterized by spatially coordinated expression of ligands and receptors across immune, tumor, and stromal compartments at TLS-tumor interfaces.

Association between TLS maturation and tumor genomic and molecular subtypes

We next evaluated TLS maturation states in three cancer types (CRC, LIHC, and BLCA) with rich clinical and molecular annotations (Fig. 5).

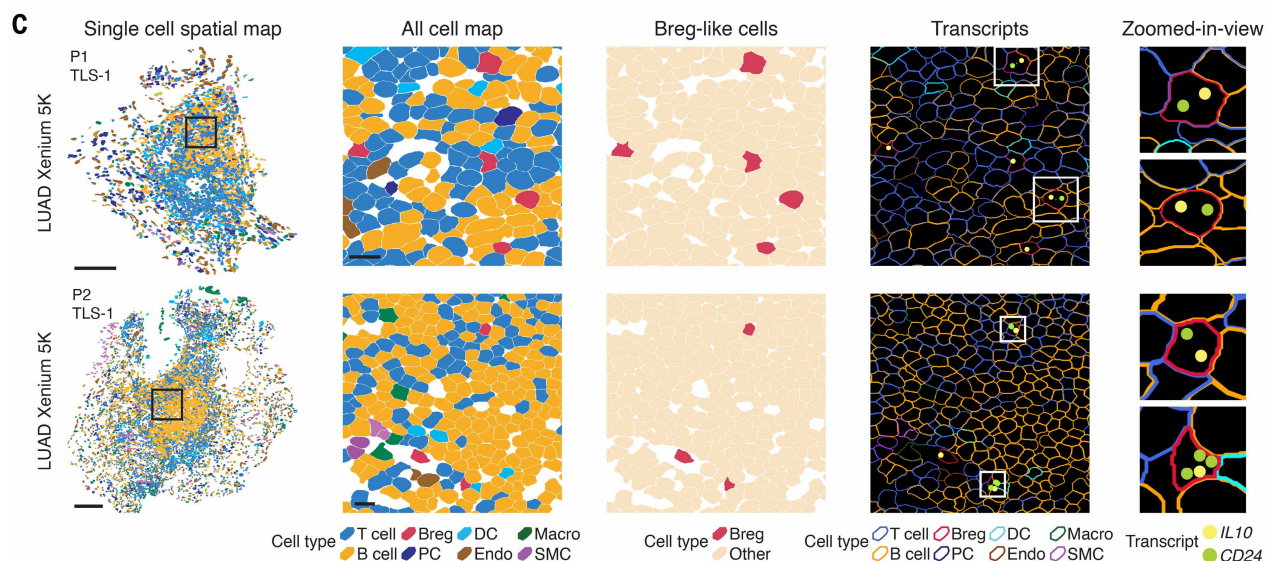


Fig. 3. Differential cellular composition and gene programs across TLS maturation states. (A) Heatmaps showing differential expression of key lineage markers, cell states, signaling pathways, cytokines, and chemokines across E-TLS, P-TLS, and S-TLS. Shown are markers for major TLS components, including B cell and T cell subsets, proliferative programs, and immunoregulatory cytokines and chemokines. Bn, naïve B cell; Bm, memory B cell; Tn, naïve T cell; Tcm, central memory T cell; Teff, effector T cell; Tfh, T follicular helper cell; Treg, regulatory T cell. (B) PC dissemination and antibody isotype distribution across TLS maturation states in ccRCC. TLSs were identified by PNAid, CD20, CD23, and BCL6 and outlined with yellow dashed lines. White squares indicate regions shown at higher magnification in E-TLS (top), P-TLS (middle), and S-TLS (bottom). Scale bar, 100 μ m. (C) Spatial mapping of IL10⁺ B cells (Breg-like) within TLSs using Xenium 5K in lung adenocarcinoma. Two representative Xenium 5K ST fields are shown. From left to right: whole-TLS single-cell maps with annotated cell types. Corresponding zoomed-in regions highlight Breg-like cells within mature TLSs; maps show *IL10* coexpression with *CD24* and/or *CD1D* within individual B cells; and higher-magnification views highlight transcripts of *IL10* and *CD24* within Breg-like cells. Scale bars, 100 μ m (left) and 10 μ m (middle).

In the CRC cohort (36 samples from 23 patients) from Heiser *et al.* (38) (Fig. 5A), TLSs were more prevalent in right-sided tumors (Fig. 5B), consistent with a more immunogenic microenvironment (39). E-TLSs appeared more frequently in adenoma regions, whereas S-TLSs were predominantly observed within carcinoma regions (Fig. 5, C and D). S-TLSs showed a trend toward enrichment in higher-grade tumors, with their proportions progressively increasing from G1 to G3 ($P = 0.058$, Pearson's χ^2 test) (Fig. 5E); were exclusively observed in *KRAS* wild-type (WT) tumors (Fig. 5F, left), and were more frequent in patients with metastatic disease (Fig. 5F, right).

In the LIHC cohort (21 samples from seven patients) from Wu *et al.* (40) (Fig. 5G), TLSs were most prevalent at the tumor leading edge compared with in tumor cores (Fig. 5H). E-TLSs and S-TLSs were more common at the tumor boundaries, whereas P-TLSs appeared to be enriched within the tumor (Fig. 5, I and J). Notably, S-TLSs were observed predominantly in hepatitis B-positive (HBV⁺) patients (Fig. 5, H, K, and L), consistent with prior reports that chronic inflammation associated with HBV infection may facilitate TLS maturation within the liver (41). More advanced disease stages (Barcelona Clinic Liver Cancer stage B/C) also tended to show higher prevalence of mature TLSs (Fig. 5L).

In our in-house BLCA cohort (10 patients: 5 MTAP⁺ and 5 MTAP⁻; MTAP, methylthioadenosine phosphorylase) (Fig. 5, M and N), S-TLSs were not detected. P-TLSs were more frequently located in stromal regions, whereas E-TLSs predominated within tumor regions (Fig. 5, O and P), a pattern distinct from CRC and LIHC cohorts. Consistent with prior reports linking MTAP loss to an immune-cold TME (42), MTAP⁻ tumors showed a trend toward a higher prevalence of TLS absence and fewer P-TLSs compared with that MTAP⁺ tumors (Fig. 5Q). E-TLSs also tended to be enriched in chemotherapy nonresponders (Fig. 5Q).

Overall, across cohorts, we observed consistent but not statistically powered trends linking TLS maturation states with tumor pathological grade, genomic context, and chronic inflammatory etiologies. Across cohorts, mature TLSs were often localized to PT or stromal regions rather than tumor cores. The observation of mature TLSs in more aggressive or advanced-stage tumors may be consistent with an immune response emerging alongside tumor progression rather than indicating a uniformly protective role. Collectively, TLS maturation state and spatial context may serve as indicators of immune engagement or dysfunction across clinical settings; however, current ST cohorts remain small and heterogeneous, limiting statistical power, and larger, subtype-resolved cohorts will be needed for definitive pan-cancer inference.

Scalable AI framework for TLS detection and maturation classification from H&E images

Given the limited size of available ST cohorts, we developed an AI-enabled framework for automatic TLS detection and maturation classification using H&E-stained WSIs alone. We curated a total of 3071 WSIs from 10 cohorts, including 6 TCGA cohorts [lung squamous cell carcinoma (LUSC, $n = 374$), BLCA ($n = 345$), KIRC ($n = 299$), LUAD ($n = 540$), STAD ($n = 442$), and COAD ($n = 458$)] and 4 independent therapy cohorts [ovarian clear cell carcinoma (OCCC, $n = 39$) (43), high-grade serous

ovarian carcinoma (HGSOC, $n = 286$) (44), and BRCA ($n = 203$, $n = 85$) (45, 46)] (Fig. 6A, left). To achieve this, we first applied HookNet-TLS (47) to detect TLSs in WSIs, followed by segmentation and TLS image extraction, yielding a total of 25,088 labeled TLSs across the 10 cohorts (Fig. 6A, middle). We then leveraged YOLOv8 (48), an advanced convolutional neural network for object detection and classification, to train a two-step classifier that first distinguishes E-TLSs from mature TLSs (mTLSs), followed by subclassification of mTLSs into P-TLSs and S-TLSs (Fig. 6A, right).

Training data were derived from three TCGA cohorts (LUSC, BLCA, KIRC), comprising 956 TLS images automatically segmented from H&E WSIs. TLS images were split into 80% training and 20% held-out test sets, stratified by cancer type, with manual annotations from experienced pathologists used as ground truth (fig. S5A; see materials and methods). To mitigate batch effects arising from variability in staining, scanning, and tissue preparation, YOLOv8 was trained with extensive data augmentation (e.g., flips, rotations, scaling, cutout, mix-up, brightness and contrast shifts, hue and saturation jitter, and Gaussian blur and noise). This augmentation strategy was designed to improve robustness to intersite and interplatform variability, enabling reliable TLS classification across heterogeneous histopathology datasets. The classifiers demonstrated robust performance across cancer types, with E-TLS versus mTLS and P-TLS versus S-TLS classification accuracies of 95.1 and 85.0% in BLCA, 90.0 and 95.6% in LUSC, and 84.0 and 89.5% in KIRC, respectively (figs. S5, B to E).

To assess cross-cancer generalizability, we applied the trained classifier directly to the 695 segmented TLS images extracted from H&E WSIs in our spatial discovery cohorts (Fig. 1A), using TLS maturation states derived from matched ST data as the reference standard (Fig. 1B). The model recapitulated the overall TLS-state distribution (fig. S6A) and achieved high accuracy for both the E-TLS versus mTLS step and the P-TLS versus S-TLS subclassification step (fig. S6, B and C). Specifically, the model correctly identified 92.3% of E-TLSs and 90.2% of mTLSs [overall accuracy 91.4%; 95% confidence interval (CI) 89.0–93.3; receiver operating characteristic area under the curve (ROC AUC) 0.908; fig. S6B]. For subclassification of mTLSs, the model correctly identified 98.0% of P-TLSs and 87.5% of S-TLSs (fig. S6C), with an overall accuracy of 94.7% (95% CI 91.4–97.0) and an ROC AUC of 0.928, despite class imbalance. Notably, this performance was achieved in multicancer ST cohorts not represented during training, without any retraining or parameter adjustment. Together, these results demonstrate that the YOLOv8-based classifier generalizes robustly across cancer types and tissue contexts.

TLS maturation varies with tumor stage and tumor-immune genomic context

Next, we applied the AI-enabled TLS detection and maturation framework to H&E WSIs from all 6 TCGA cohorts (Fig. 6, A and B) to evaluate whether the stage-associated TLS maturation trends observed in our spatial discovery cohorts (Fig. 5) were reproducible at scale. Across STAD, COAD [microsatellite instability low (MSI-L) and microsatellite stable (MSS)], and KIRC, we observed a stage-associated shift in TLS composition, with increased proportions of P-TLSs and S-TLSs in more advanced-stage tumors relative to those in early-disease (Fig. 6C). By

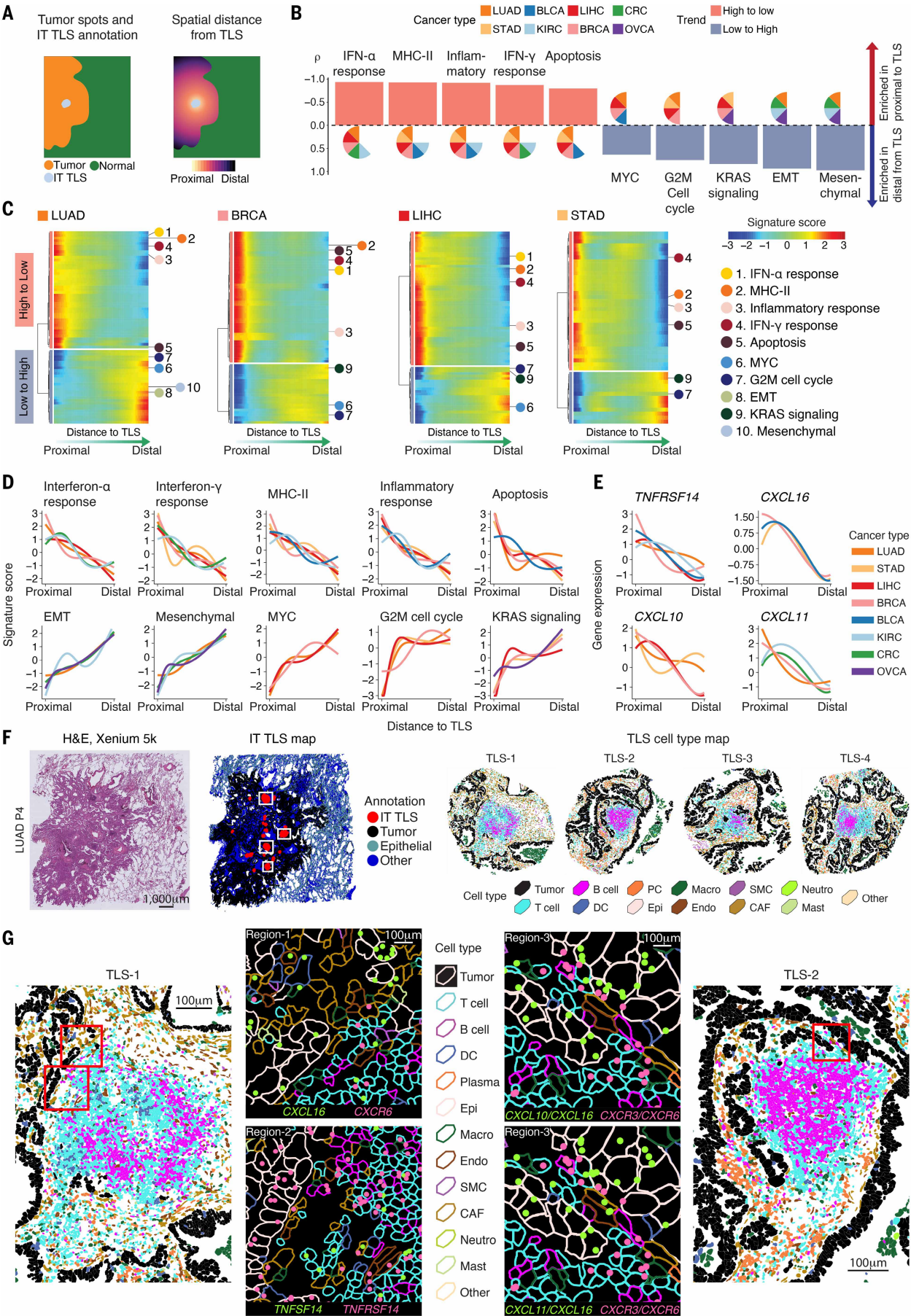


Fig. 4. Spatial dynamics of tumor pathway activity relative to TLS proximity. (A) Schematic illustrating an IT TLS (left) and how the distance between its center and nearby tumor regions is measured (right). (B) Bar plots showing correlations between the top 10 enriched pathways in tumor regions and their distance from IT TLS. “High-to-low” pathways exhibit high expression near TLS that decreases with distance, whereas “low-to-high” pathways show the opposite trend. Pie charts indicate cancer types associated with each enrichment trend. (C) Heat maps displaying spatial expression dynamics of pathway scores across tumor regions ordered by proximity to IT TLS (left, proximal; right, distal) in four representative cancer types. Pathways are hierarchically clustered into “high-to-low” (top) and “low-to-high” (bottom) groups. Numbered and color-coded annotations indicate the top 10 pathways. (D) Line plots showing expression dynamics of the top 10 pathways [from (C)] across tumor distance from IT TLS in eight cancer types. Each line represents a different cancer type. (E) Representative examples of distance-dependent expression changes for TLS-associated cytokines and chemokines (*TNFSF14*, *CXCL11*, *CXCL16*, *CXCL10*) across eight cancer types. (F) Representative spatial maps from a Xenium tumor sample (left to right): whole-slide H&E image, IT TLS map, and zoomed-in spatial cell type maps for four representative TLSs (1 to 4), showing distinct immune and stromal compartments with major cell types annotated. (G) Representative Xenium ST map highlighting selected TLS-1 (left) and TLS-2 (right) from (F) and higher-magnification views of boxed regions from each TLS (middle), displaying cell boundaries overlaid with transcript expression for TLS-associated chemokines (*TNFSF14*, *CXCL16*, *CXCR6*, *CXCL10*, *CXCL11*, *CXCR3*). For (D) and (E), the x axes show distance from IT TLS (proximal to distal), and the y axes show normalized signature score (z-score).

contrast, LUSC, LUAD, BLCA, and COAD [microsatellite instability high (MSI-H)] did not exhibit a consistent pattern (fig. S7A). When cohorts were combined, TLS maturation-state distributions shifted significantly across stages I to III, whereas stage IV tumors showed substantial overlap and no further separation (Fig. 6D). Overall, TLS maturation frequently, but not uniformly, tracks with tumor stage, with the association most evident in early-to-intermediate stages.

We further assessed how TLS maturation states relate to established immune and genomic features defined in prior TCGA analyses (fig. S7B) (49). We observed a progressive increase from E-TLSs to P-TLSs and then to S-TLSs in their associations with tumor mutation burden (including single-nucleotide variant and indel neoantigens), homologous recombination deficiency, intratumoral heterogeneity, lymphocyte infiltration, leukocyte fraction, IFN- γ response, and wound-healing signals (fig. S7B). These patterns are consistent with a maturation-associated trajectory in which mature TLSs occur in tumors with more genomically altered and immune-active microenvironments. We note that these associations may be influenced by limitations inherent in bulk transcriptomic deconvolution and therefore warrant cautious interpretation.

TLS maturation-aware composite scoring enables robust patient stratification

We next evaluated the prognostic performance of TLS maturation using commonly used patient classification strategies (21–24), including TLS presence versus absence (Fig. 6E, left, and fig. S7C), dominant TLS maturation state (Fig. 6E, middle, and fig. S7D), and broader groupings of immature (i.e., E-TLS) versus mature TLSs (Fig. 6E, right, and fig. S7E). Across six cancer types, these approaches failed to provide consistent prognostic separation. Binary classification based on TLS presence alone did not reliably stratify overall survival (OS) in Kaplan-Meier (KM) analyses, regardless of TLS maturation state. Similarly, classification by dominant TLS state using fine-grained or grouped categories did not yield robust survival differences across cohorts. Only KIRC showed a significant association with S-TLS dominance, underscoring the cancer-type specificity of TLS-outcome relationships rather than a universal rule (Fig. 6E and fig. S7, C to E). Together, these results highlight fundamental limitations of existing TLS-based stratification strategies.

Building on insights from spatial TLS profiling, we sought to develop a patient stratification framework that captures both TLS abundance and maturation state heterogeneity within a tumor rather than collapsing tumors into a single dominant state. To this end, we derived a data-driven composite TLS score using principal component analysis (PCA) on log-transformed, z-score-normalized counts of E-TLSs, P-TLSs, and S-TLSs (Fig. 6F; see materials and methods). This unsupervised approach captures the dominant axes of variation in TLS state composition across tumors, yielding a continuous score that reflects both TLS abundance and maturation state diversity within individual tumors. Applying a median-based threshold, we stratified TLS-positive tumors into two groups: C1 and C2 (Fig. 6F and table S7). Among TLS+

samples, the C1 group was enriched for P-TLSs and S-TLSs, whereas the C2 group was dominated by E-TLSs (fig. S8A and table S8), explicitly accounting for intratumoral TLS heterogeneity.

We then assessed the prognostic significance of this TLS maturation-aware composite scoring approach (Fig. 6, F and G). Notably, patients in the C1 group exhibited significantly improved OS compared with those in the C2 group, and this association was consistently observed across all six cohorts, with the TLS composite score-based framework outperforming conventional stratification methods in every cohort examined (Fig. 6G and fig. S7, C and E). Notably, multivariable Cox proportional hazards models adjusting for tumor stage confirmed the independent prognostic value of the C1 group (fig. S8B), indicating that integrating TLS maturation composition yields prognostic information beyond maturation state alone. Together, these results support TLS maturation-aware compositional profiling as a robust prognostic stratifier beyond conventional TLS metrics and tumor stage.

To examine whether TLS maturation is associated with therapeutic response, we first analyzed an independent in-house cohort of OCCC patients treated with ICB (Fig. 6H) (43). Consistent with prior observations linking *PPP2R1A* loss-of-function mutations to improved OS following immunotherapy in OCCC patients, *PPP2R1A*-mutant tumors exhibited significantly higher counts of P-TLSs and S-TLSs compared with *PPP2R1A* WT tumors (Fig. 6I, top). TLS maturation state composition prior to treatment also differed significantly between short- and long-survival groups, with enrichment of mature TLS states in long-survival cases (Fig. 6I, bottom). Notably, these differences were not significant at the on-treatment timepoint, suggesting that baseline TLS composite scores may be more informative of therapeutic outcome than on-treatment changes. Similar directional trends were observed when stratifying patients by treatment response, although statistical power was limited (fig. S7F).

Lastly, we evaluated whether associations between TLS maturation and therapeutic response extend to nonimmunotherapy treatment settings, including chemotherapy and targeted therapy, where treatment exposure is much more common clinically. Across three independent cohorts spanning combination chemotherapy plus antiangiogenic therapy (HGSOC, Fig. 6J) (44), neoadjuvant combination chemotherapy (breast cancer, Fig. 6L) (45), and anti-HER2 targeted therapy (breast cancer, Fig. 6N) (46), enrichment of mature TLS states, particularly P-TLSs, was consistently associated with favorable treatment response. In each cohort, responders exhibited higher P-TLS and/or S-TLS abundance, increased fractions of mature TLS states, and preferential classification into the C1 group (Fig. 6, J to O). Despite differences in tumor type and treatment modality, these results were directionally concordant, supporting the general relevance of TLS maturation composition to therapeutic response across diverse clinical contexts.

Discussion

TLSs represent specialized immune niches that can shape antitumor immunity across solid tumors (3, 40, 50–52). Yet their spatial organization, maturation states, relationships to tumor cells and the surrounding

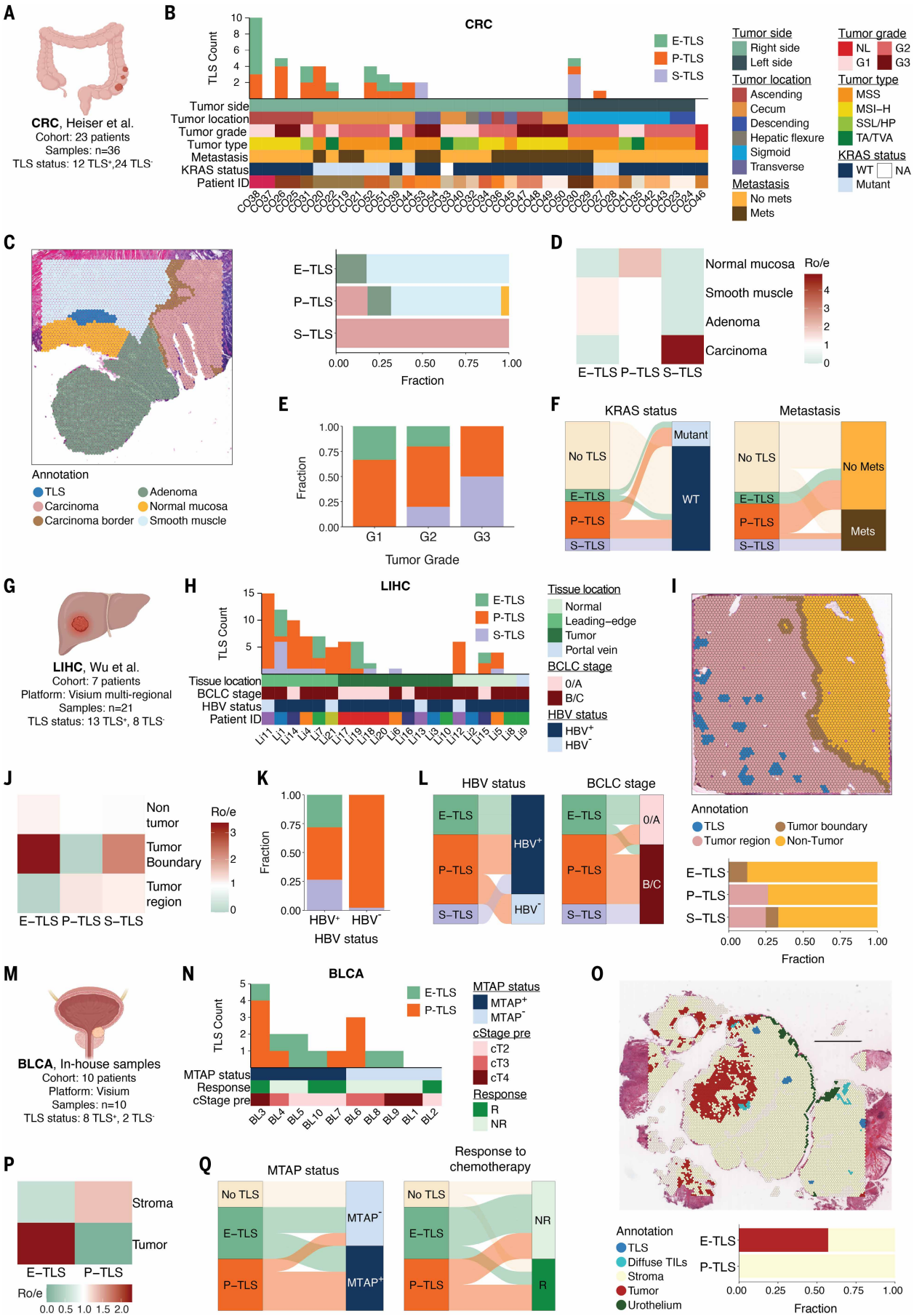


Fig. 5. Association between TLS maturation states, tumor features, and driver mutation status across cancer cohorts. (A to F) Colorectal cancer cohort (38). (A) Cohort overview. (B) TLS count and maturation states across patient samples (column) annotated with clinical and molecular features. SSL: sessile serrated lesion, HP: hyperplastic polyp, TA: tubular adenoma, TVA: tubulovillous adenoma, Mets: Metastasis, No Mets: No metastasis. (C) Pathology annotation of tissue compartments from the original study (38). The bar plot shows the fractions of tissue compartments colocalized with TLSs by maturation state. (D) Heatmap depicting the relative enrichment of tissue compartments across TLS maturation states ($R_{o/e}$ values). (E) Stacked bar plot showing the relative fractions of E-TLS, P-TLS, and S-TLS by tumor grade (G1 to G3). (F) Enrichment of TLS maturation states by *KRAS* mutation status (left) and metastases (right). **(G to L)** Primary liver cancer cohort (40). (G) Cohort overview. (H) TLS count and maturation states across patient samples (column), annotated with clinical and molecular features. [(I) and (J)] Tissue region annotations (tumor bed, boundary, nontumor; see materials and methods) [(I), top]] and TLS distribution across regions by maturation state, shown as a bar plot [(I), bottom]] and an $R_{o/e}$ enrichment plot (J). (K) TLS maturation state distribution by HBV status. (L) Enrichment of TLS states by HBV infection status (left) and tumor stage (right). **(M to Q)** In-house BLCA cohort. (M) Cohort overview. (N) TLS count and maturation state distribution across samples (column), annotated with clinical and molecular features, including tumor stage (cStage pre), *MTAP* status, and treatment response. R, responder; NR, nonresponder. [(O) and (P)] Tissue compartments annotated by board-certified pathologists [(O), top]. TILs, tumor-infiltrating lymphocytes. TLS distribution between tumor and stroma by maturation state, shown as a bar plot [(O), bottom]] and an $R_{o/e}$ enrichment plot (P). (Q) Enrichment of TLS maturation states by status of *MTAP* (left) and chemotherapy response (right). *MTAP*⁺, *MTAP* WT; *MTAP*⁻, *MTAP* loss. For panels (C), (I), and (O), TLSs were annotated using our own workflow. For panels (D), (J), and (P), relative enrichment was calculated using $R_{o/e}$ (see materials and methods). For panels (F), (L), and (Q), proportion tests did not show significant differences in TLS maturation state distributions across groups.

TME, and clinical relevance have not been systematically characterized at scale across cancer types. In this study, we used ST data to build a pan-cancer atlas of TLSs across 12 cancer types, with ultrahigh-plex, single-cell-resolved spatial RNA and protein profiling providing orthogonal validation in selected cohorts and integrated these findings with histopathology and clinical data. By developing a scalable AI-enabled framework to detect and classify TLSs directly from routine H&E WSIs, we have extended TLS analysis from limited spatial datasets to thousands of tumors. Leveraging this scalability, we have introduced a TLS maturation-aware composite scoring strategy that captures within-tumor TLS state composition and enables robust patient stratification across cancers and treatment contexts. Together, this work establishes generalizable and clinically scalable frameworks for TLS profiling and highlights TLS state composition as a key dimension of tumor immune organization with translational relevance. It also provides a foundation for prospective evaluation of TLS-informed biomarkers in clinical settings.

Across cancer types, TLSs were neither spatially uniform nor immunologically equivalent. TLS localization and maturation states varied substantially by tumor type and tissue context, with TLSs distributed across IT, PT, and DT regions in tumor type-specific patterns. Consistent with this view, in ST cohorts with richer clinical annotations, TLS prevalence and maturation also showed associations with tumor-intrinsic and etiologic contexts (e.g., *KRAS* status in CRC, HBV status in LIHC, and *MTAP* loss in BLCA) (38, 40, 42), suggesting that tumor genetics, tissue context, and inflammatory milieu can differentially support or constrain TLS formation and maturation. Although these observations are based on relatively small cohorts, they motivate larger, subtype-resolved studies to more definitively link TLS evolution to tumor biology. Applying the AI framework at scale further revealed that TLS maturation frequently, but not uniformly, tracked with tumor stage, with progressive enrichment of more mature TLS states in early-to-intermediate stages and attenuation of this association in advanced stages. Together, these patterns suggest that TLS maturation reflects dynamic interactions between tumor progression and local immune organization rather than a fixed protective phenotype.

Our analyses revealed that TLS maturation is accompanied by coordinated multicompartment niche remodeling within the TME. TLSs follow a progressive trajectory from E-TLSs to P-TLSs and ultimately S-TLSs (1, 16), marked by coordinated enrichment of immune and stromal programs. As TLSs matured, we observed increased representation of activated T and B cell populations, specialized effector and regulatory subsets (e.g., Breg-like cells), and chemokine and cytokine networks that support structured B cell-T cell interactions (9, 28). These changes likely facilitate compartmentalized immune activation while concurrently engaging immunoregulatory programs that contribute to local immune homeostasis (27, 35, 36, 53–56). In parallel, TLS maturation involves progressive engagement of antigen-presenting cells (57–59), shifts in macrophage phenotypes (60–65), and increasing

specialization of endothelial cells and fibroblasts (65–67), together with vascular remodeling that facilitates immune cell recruitment and retention, and spatial organization. These multicompartmental changes are further reinforced by stress-adaptive transcriptional programs, including heat-shock and IFN-stimulated responses, which may help maintain TLS integrity and function under inflammatory and metabolic stress. Collectively, these features suggest that mature TLSs are not simply “more immune” but that they represent stabilized, self-supporting niches that can sustain local immune organization under inflammatory and metabolic stress (68–70).

Although prior studies have implicated TLSs in regulating local immune responses (1, 5, 9, 16, 29, 36), our pan-cancer analysis provides a systematic view that links TLSs to the spatial logic of immune-tumor patterning within the TME. Rather than passive byproducts of immune infiltration, TLSs appear to both mark (and potentially help maintain) spatially compartmentalized immune activation at tumor interfaces. We found that TLS proximity was associated with consistent spatial gradients across cancers, with immune activation enriched in TLS-proximal tumor regions and proliferative and invasive programs enriched distally (23). We further identified ligand-receptor axes with established immunological roles that align with these proinflammatory gradients. The TNFSF14-TNFRSF14 (LIGHT-HVEM) axis regulates local immune activation and has been shown to enhance antitumor immunity (71). The CXCL10/11-CXCR3 axis is a canonical pathway for effector cell recruitment, consistent with inflammatory signaling observed in TLS-proximal tumor regions (72, 73), whereas the CXCL16-CXCR6 axis has been implicated in chronic inflammation and effector-cell retention (74). Collectively, these findings support a model in which TLSs are embedded within short-range cytokine and chemokine fields that promote immune cell recruitment and retention and reinforce inflammatory signaling at TLS-tumor interfaces. The distance-dependent immune gradients that we observed around IT TLSs may reflect multiple, nonmutually exclusive processes, including (i) localized cytokine or chemokine fields produced by TLS-associated immune cells, (ii) composition shifts driven by immune cell redistribution or retention near TLS borders, and (iii) localized transcriptional reprogramming of adjacent tumor and stromal cells, consistent with the spatially coordinated ligand-receptor patterns mapped at single-cell and subcellular resolution. These spatial associations are correlative; establishing causality will require functional perturbation and longitudinal, higher-resolution spatial profiling.

To evaluate clinical relevance, we applied our AI-based TLS framework at scale and introduced a stratification strategy based on TLS compositional maturation profiles. By summarizing within-tumor TLS state composition into a continuous score, this approach consistently outperformed conventional TLS metrics (21–24) in prognostic prediction across cancer types. Notably, the C1 group was associated with significantly improved OS independent of tumor stage, indicating that

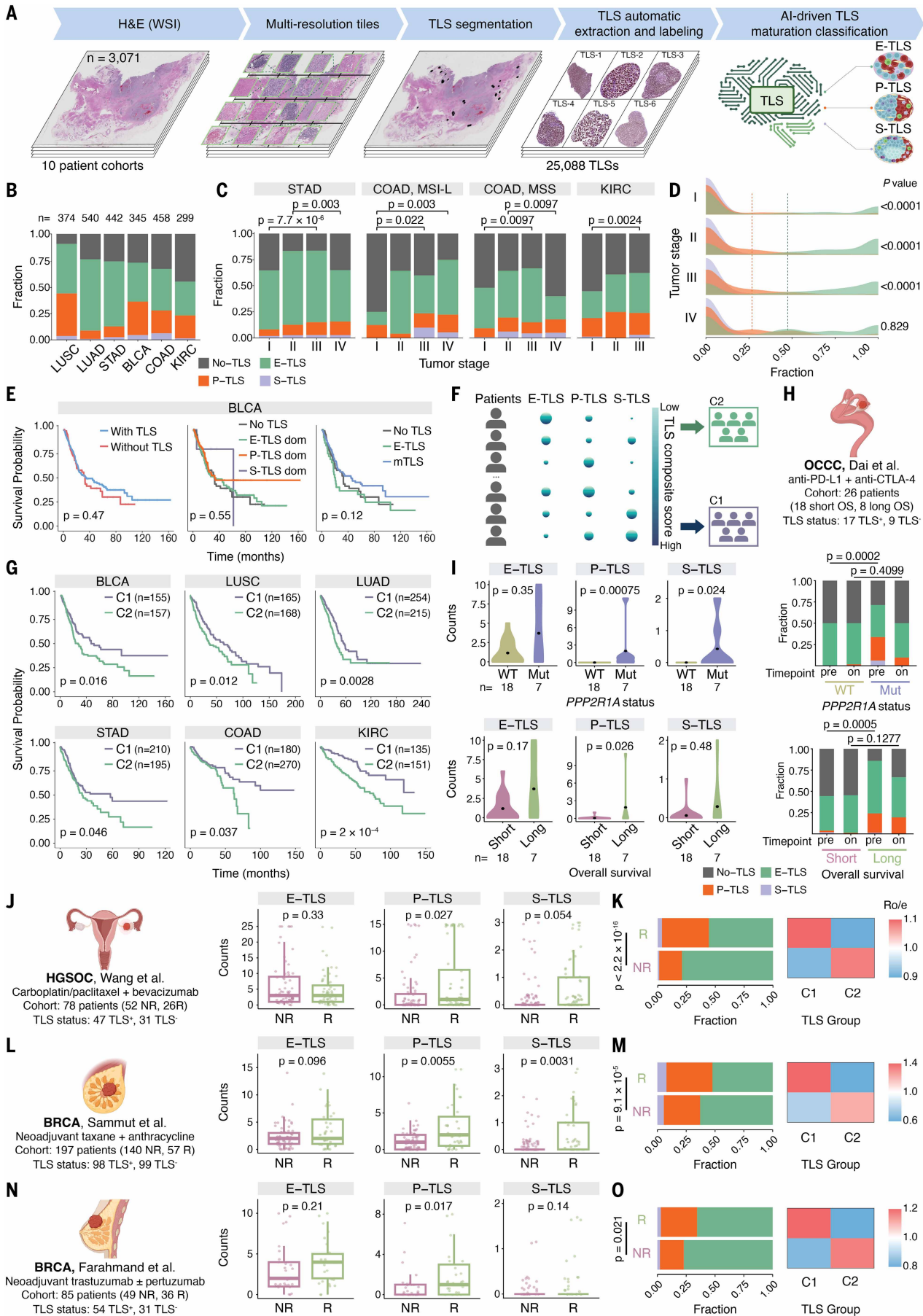


Fig. 6. AI-driven TLS classification and its prognostic relevance in TCGA cohorts. (A) Schematic overview of the AI-based TLS detection and classification frameworks applied to 3071 WSIs from 10 patient cohorts [TCGA: LUSC, KIRC, BLCA, LUAD, STAD, COAD; therapy cohorts: OCCC, HGSO, and two BRCA cohorts (45, 46)]. Each WSI was tiled at multiple resolutions for analysis. TLSs were detected using a deep learning model (HookNet-TLS) (47), then segmented and automatically labeled using our custom workflow, yielding 25,088 individual TLS images. An AI model (YOLOv8) (48) was trained and then applied to classify TLSs into three maturation states: E-TLS, P-TLS, and S-TLS (see materials and methods) (48). (B and C) Bar graphs display the relative fractions of TLS maturation states across the six cancer types (B) and those stratified by AJCC tumor stages across STAD, COAD (MSI-L, MSS), and KIRC (C). (D) Distributions of E-TLS, P-TLS, and S-TLS fractions across tumor stages. (E) Kaplan-Meier curves for overall survival stratified by conventional TLS classification based on the presence versus absence of TLSs (left) or by dominant TLS maturation state categorized into four (middle) or three groups (right). mTLS, mature TLSs. (F) Schematic of the TLS composite score derived from log1P-transformed counts of E-TLS, P-TLS, and S-TLS per sample. Z-score-normalized TLS counts were subjected to PCA, and a weighted sum of PC1 and PC2 (proportional to explained variance) was used as the composite score, capturing TLS abundance and maturation composition. Patients were stratified into C1 and C2 groups based on the median TLS composite score. (G) KM survival curves comparing overall survival between two patient groups stratified by our new approach (C1 and C2). (H) Cohort overview (43). (I) Top row (stratified by PPP2R1A mutation status), from left to right: violin plots comparing per-patient TLS counts by maturation state and TLS maturation-state composition comparisons. Bottom row (stratified by overall survival group), from left to right: violin plots comparing per-patient TLS counts by maturation state and TLS maturation-state composition comparisons. Short survival was defined as death before the KM 75th percentile time, whereas long survival was defined as survival at or beyond that threshold. Twenty-five patients had pretreatment samples, and 14 had on-treatment samples; analyses in the left panel were performed using the pretreatment samples. (J and K) HGSO cohort (44). (L and M) Breast cancer cohort (45). (N and O) Invasive breast carcinoma cohort (46). [(J), (L), (N)] (Left) Cohort overview. (Right) Boxplots of per-patient counts of E-TLS, P-TLS, and S-TLS stratified by response status. [(K), (M), (O)] (Left) Fraction of TLS maturation states stratified by response status. [(K), (M), (O)] (Right) Heatmap of $R_{o/e}$ of TLS groups (C1 versus C2) by response status. *P*-values were from log-rank tests [(E) and (G)], two-sided Wilcoxon rank sum tests [(I), left; (J); (L); and (N)] and χ^2 tests [(C); (D); (I), right; (K); (M); (O)].

TLS composition provides prognostic information beyond maturation state alone. These findings highlight a key limitation of existing TLS stratification strategies, which often collapse complex intratumoral TLS heterogeneity into binary or single-label classifications. In practice, individual tumors often harbor multiple TLSs spanning distinct maturation states, a feature that is lost when tumors are reduced to a single dominant category. Conceptually, these results argue that TLS “quality” is not captured by presence or a single dominant state but by the mixture and abundance of coexisting TLS states within a tumor, an axis of intratumoral immune organization that prior binary schemes erase. To our knowledge, this represents the first AI-enabled, proportion-aware TLS stratification framework that enables large-scale TLS phenotyping and prognostic stratification directly from routine H&E images alone. Using this maturation-aware composite score, we demonstrate robust and reproducible associations with OS across independent cohorts.

Extending these survival associations, enrichment of mature TLS states in responders across immunotherapy, chemotherapy, and targeted therapy cohorts suggests that TLS maturation may reflect a broader immune-competent tissue state, not a feature restricted to immunotherapy response alone. Although chemotherapy, targeted therapy, and antiangiogenic agents do not directly target immune checkpoints, they can reshape the tumor immune microenvironment through immunogenic tumor cell death, antigen release, Fc-mediated effector mechanisms, and vascular or stromal remodeling. Mature TLSs are also associated with PC differentiation and antibody production, potentially supporting antibody-mediated effector functions (e.g., antibody-dependent cellular cytotoxicity and complement activation). In addition, their frequent colocalization with specialized vasculature (e.g., HEVs) may facilitate effector recruitment and local immune amplification. Together, these observations provide a plausible basis linking mature TLS enrichment to improved response and survival, although direct mechanistic testing is needed.

Collectively, these findings position TLS compositional maturation as a clinically scalable spatial phenotype and provide an AI-enabled path to quantify it from routine histology. Prospective studies should test whether TLS composition improves risk and response modeling beyond established clinicopathologic and molecular predictors, and whether TLS-informed stratification can guide clinical trial design or therapeutic modulation strategies.

Limitations of the study

This study has several limitations. First, most ST datasets were generated using spot-level ST platforms. We applied resolution-enhancement approaches, orthogonal validation using single-cell ST and SP technologies, and spatial multimodal integration. However, current ST and SP platforms

still lack validated reagents for isotype- and subclass-resolved plasma-cell profiling. As spatial technologies advance, systematic cross-cancer dissection of TLS-associated antibody output, including antibody isotypes and subclasses and their clinical associations, will be an important direction for future work. Second, longitudinal spatial profiling will be essential to define how TLS maturation and compositional states evolve during tumor progression and in response to therapy. Third, available ST cohorts remain small and heterogeneous with variable clinical annotation; large, subtype-resolved datasets will be needed for definitive pan-cancer inference. Lastly, our analyses are correlative, and causal mechanisms governing TLS formation, maturation arrest, and immune function will require experimental perturbation in appropriate model systems. More broadly, TLSs are three-dimensional (3D) structures, and section-level sampling remains a limitation of current ST approaches; future studies integrating serial sections and nondisruptive 3D spatial profiling will be important to resolve TLS architecture in situ.

Materials and methods

Human specimens

All human data presented in this study are de-identified to protect patient privacy. All human tumor tissues collected at The University of Texas MD Anderson Cancer Center were obtained from patients who provided signed informed consent and were under institutional review board (IRB)-approved protocols.

Pancreatic adenocarcinoma (PAAD)

We collected Visium ST datasets from two cohorts of PAAD samples. The first cohort comprises 13 patients who were initially diagnosed with pancreatic ductal adenocarcinoma (PDAC) and underwent a “warm” research autopsy within four hours post-mortem, conducted under the University of Nebraska Medical Center (UNMC) Rapid Autopsy Program under a protocol approved by their institutional review board (IRB #091-01). In total, 13 snap-frozen primary tumor tissue samples were obtained from these rapid autopsies. Written informed consent was obtained from all patients. Details of tissue collection, processing, and ST data generation have been described in our recent study by Pei *et al.* (75). The second cohort comprises 10 patients who were initially diagnosed with PDAC and were treatment-naïve at the time of sample collection. Details of tissue collection, processing, and ST data generation have been described in our recent study by Liu *et al.* (14). Written informed consent was obtained from all patients. Treatment-naïve tissues were collected under an IRB-approved protocol (PA15-0014) at The University of Texas MD Anderson Cancer Center.

BLCA

Ten patients with primary muscle-invasive bladder carcinoma (T2–T4) were evaluated at The University of Texas MD Anderson Cancer Center and underwent transurethral resection of bladder tumor. All patients were treatment-naïve at the time of surgery. Formalin-fixed paraffin-embedded (FFPE) BLCA tumor tissues in this study were obtained from the Department of Pathology at The University of Texas MD Anderson Cancer Center under institutional IRB-approved research protocol PA17-0577. Details of tissue collection, processing, and ST data generation have been described in our recent study by Jiang *et al.* (76).

LUAD

Study participants were evaluated at Juntendo University (Juntendo, Japan). Specimens were approved for study by Institutional Review Boards and according to the international ethical guidelines for biomedical research involving human subjects (Council for International Organizations of Medical Sciences). Informed written consents were received from all subjects as required. Following surgery, resected tissues were processed by an experienced pathologist assistant for FFPE, followed by sectioning and staining with H&E.

Collection of ST datasets

ST data for the discovery cohort, consisting of 340 samples from 255 individuals across 12 cancer types (Fig. 1A and tables S1 and S2), were collected from both in-house and public sources. In-house data consisted of 74 samples (21.8%) generated at the Advanced Technology Genomics Core (ATGC) facility at the University of Texas MD Anderson Cancer Center, spanning PAAD, BLCA, STAD, and LUAD. Public data comprised the remaining 266 samples (78.2%), obtained from previously published studies and from the 10x Genomics website (3, 38, 40, 51, 76–92). Dataset origins and accession numbers are summarized in table S1, with in-house samples labeled as “in-house” in the Data Type column. Table S2 further details sample type (Visium/ST), tissue preparation (FFPE or Fresh Frozen), and TLS presence. All datasets, regardless of origin, were processed and analyzed using the same standardized pipelines in downstream analyses.

Visium ST data generation

ST was performed using the 10x Genomics Visium platform following manufacturer's protocols tailored to tissue preservation type. Fresh frozen (FF) PAAD and FFPE (BLCA, LUAD, STAD) samples were sectioned (10 μ m for FF tissues; 5 μ m for FFPE tissues), methanol-fixed (PAAD only), H&E-stained, and imaged at high resolution prior to library preparation. RNA quality was assessed using the Agilent TapeStation to determine RNA Integrity Number (RIN) for FF tissues (extracted with the RNeasy Mini Kit, Qiagen), or the Agilent RNA 6000 Pico Kit to determine DV200 for FFPE tissues (extracted with the RNeasy FFPE Kit, Qiagen). Libraries were prepared using Visium kits, direct slide capture for FF samples and CytAssist-enabled workflows for FFPE tissues, and were sequenced on Illumina NovaSeq platforms.

Details of tissue-specific protocols are summarized below: PAAD samples were processed on standard Visium slides (6.5 $\times$ 6.5 mm), with RNA quality assessed via TapeStation RIN, methanol fixation prior to staining, and a 15-min permeabilization step. BLCA, LUAD, and STAD samples were processed using the Visium CytAssist system with DV200 > 30%, 11 $\times$ 11 mm regions of interest (ROIs). Libraries were sequenced to $\geq 50,000$ read pairs per spot. Details regarding tissue collection, processing, and ST data generation are provided in our recent studies (14, 75, 76).

Automated seqIF-based spatial proteomics and imaging on COMET

Sequential immunofluorescence (seqIF) was performed using the COMET platform (Lunaphore Technologies) on FFPE tissue sections. Sections underwent baking, deparaffinization, and rehydration, followed by autofluorescence quenching and antigen retrieval (pH9 EZ AR

2 Elegance, Biogenex), following optimized in-house protocols. Slides were then processed on the COMET Stainer following the manufacturer's guidelines. Iterative cycles of immunostaining were conducted, each consisting of primary and secondary antibody incubation, image acquisition, and antibody elution, as previously described (93, 94). Antibody panels were customized for the tissue types analyzed and are listed in table S5, including antibody clone, species, dilution, acquisition channel, and staining cycle. Primary antibody incubations were performed for 8 min per round. At the completion of the staining cycles, multi-layer OME-TIFF images from each round were automatically stitched and aligned. Background fluorescence was computationally subtracted, and the images were visualized and analyzed using Lunaphore Viewer and QuPath software.

Multiplex immunofluorescence with Akoya PhenoCycler-Fusion IO60 panel

FFPE tissue sections (thickness typically 4–5 μ m) were mounted onto poly-L-lysine-coated or charged glass slides and baked to ensure adhesion prior to staining. Multiplexed cyclic immunofluorescence was performed using the Akoya PhenoCycler-Fusion (formerly CODEX) platform with the IO60 immune-oncology antibody panel. All procedures followed the manufacturer's recommended workflow for automated barcoded antibody staining and iterative imaging. Tissue sections underwent deparaffinization and antigen retrieval (citrate- or Tris-based buffer, heat-induced), followed by blocking to minimize nonspecific binding. A cocktail of DNA-barcoded primary antibodies corresponding to the IO60 panel was applied to the sections. After incubation and washing, the samples were loaded onto the PhenoCycler-Fusion system, which performs sequential hybridization and cleavage cycles. During each imaging cycle, complementary fluorescent reporters were hybridized, imaged, and stripped to enable visualization of the next set of markers. High-resolution imaging was acquired under standardized exposure settings with automated Z-stack or focal-plane optimization, as defined by the Akoya protocol. Raw images were aligned and unmixed using the PhenoCycler-Fusion software suite to correct for background and spectral overlap. Processed images were reviewed and quantified using QuPath software.

Immunofluorescence and immunohistochemistry labeling assays

FFPE human ccRCC (clear cell renal cell carcinoma) tumor and control specimens were cut into 3–4 μ m-thick sections. Human FFPE tonsil sections and/or ccRCC test tumor sections (France Tissue Bank) were used as positive controls for BCL6, CD3, CD20, CD23, MUM1 (*irf4*), IgA and IgG. Antigen retrieval was carried out on a PT-link (Dako) using the EnVision FLEX Target Retrieval Solutions at High pH (Dako, K8004) or Low pH (Dako, K8005). Endogenous peroxidase activity and non-specific Fc receptor binding were blocked with H₂O₂ 3% (Giffrer, 10603051) and Protein Block (Dako, X0909), respectively. The primary and secondary antibodies used for IHC and IF are summarized in table S6. All images were independently analyzed by three reviewers. Necrotic and serous areas were excluded from image analyses.

Multiplex immunofluorescence assay

The PNA^d/CD20/CD23/BCL6 4-plex and the IgA/IgG/MUM1 3-plex stainings were performed manually on FFPE tumor sections using a tyramide system amplification (TSA) and a conventional fluorescent-dye conjugated secondary antibody system (all secondary antibodies were diluted at 1:100). The panel combined PNA^d, CD20, CD23, and BCL6 to define TLS architecture and maturation, together with MUM1 (*IRF4*), IgA, and IgG to assess ASC identity and isotype distribution. MUM1 (*IRF4*) was used as a nuclear marker of terminally differentiated ASCs and interpreted in conjunction with IgA/IgG expression and B-lineage context to ensure specificity. Nuclei were stained with DAPI Solution (ThermoFisher, 62248) at 2 μ g/ml for 10 min. After mounting with ProLongTM Gold Antifade Mountant (ThermoFisher, P36934), the

slides were scanned at 20X magnification using a Zeiss Axio scan ZI device. Antibodies and TSA used are listed in table S6. Images were analyzed using Halo10 software (Indica labs) with appropriate cell-counting algorithms. TLS regions were identified based on CD3 (green) and CD20 (red) chromogenic staining on a consecutive slide.

TLS identification by enzymatic immunohistochemistry

Enzymatic immunohistochemical staining was performed manually to detect TLSs. Chromogenic detection was performed using HighDef red IHC chromogen (AP) (Enzo, ADI-950-140-0030) for CD20, Permanent HRP Green (Zytomed Systems, ZUC070-100) for CD3. The nuclei were counterstained with hematoxylin (Dako, S3301). After mounting with EcoMount (Biocare Medical, EM897L), the slides were scanned with a NanoZoomer (Hamamatsu). TLSs were identified using the registration module to fit one slide on the other (Halo10 software, Indica labs). Tumors were considered TLS-positive when a CD3 aggregate with CD20 staining was found with surface above $7000 \mu\text{m}^2$, containing at least 100 cells, as quantified with Halo10 software (Indica labs).

Xenium data generation

For each tissue block, H&E-stained sections were first reviewed to ensure suitable morphology, section integrity, stain uniformity, and image resolution. Criteria included the absence of folding, tearing, or detachment; appropriate stain intensity; and sufficient clarity without blurring or contrast loss. Only samples passing all quality checks proceeded to Xenium processing. For whole-tissue experiments, $11 \times 11 \text{ mm}$ regions corresponding to areas previously profiled by Visium were selected and paired on Xenium slides. Tissues were processed on the Xenium Analyzer (10x Genomics, model 1000481) according to the Xenium Prime 5K in situ gene expression protocol (CG000760). All hybridization, imaging, and decoding steps were conducted as instructed in the manufacturer's workflow.

Visium data processing and quality control

The detailed Visium data processing and quality control (QC) have been previously described in our recent studies (14, 75, 76). Briefly, the demultiplexed raw sequencing data were aligned to the human reference genome build GRCh38 (hg38) using the Space Ranger analysis pipeline (v3.0.2) from 10x Genomics with default settings to generate gene-level unique molecular identifier (UMI) count matrices. For basic filtering, spots with low-complexity libraries (e.g., $\text{nUMI} < 500$ or $\text{nFeature} < 200$) were excluded. Additionally, non-informative spots corresponding to technical artifacts or background (e.g., spots without tissue) were removed based on manual inspection using the Loupe Browser.

Gene expression enhancement

To enhance the resolution of tissue architecture, we used iStar (25), an integrative approach that combines ST with histological features. For each sample, the input data included the H&E image, spot-level gene count matrix, spatial coordinates, physical pixel size (in μm), number of pixels per spot radius, and a binary tissue mask. The H&E image was rescaled to $0.5 \times 0.5 \mu\text{m}^2$ per pixel to align its spatial resolution with the transcriptomics data. Histological features were then extracted from image patches (16×16 pixels, corresponding to $8 \times 8 \mu\text{m}^2$) centered on each ST spot. A four-layer feedforward neural network was trained to map these histological features to the corresponding spot-level gene expression using mean squared error loss. The model was trained for 400 epochs with a batch size of 128 and a learning rate of 1×10^{-4} using the Adam optimizer. After training, the model was applied across the entire tissue section to predict gene expression at the pixel level. The resulting enhanced gene expression values were normalized using min-max scaling to the [0,1] range. By achieving high-resolution mapping, iStar facilitates detailed insights into the cellular and molecular architecture of the tissue, supporting advanced analyses in tissue segmentation and TME characterization.

TLS detection, segmentation, extraction, and labeling

We applied a standardized procedure for detecting, segmenting, extracting, and labeling TLSs across all ST samples. First, we calculated the TLS score for each imputed pixel by averaging the expression levels of TLS signature genes, as detailed in table S3 and previously reported (25). This TLS score provided a quantitative measure of TLS intensity at each pixel, enabling accurate identification and spatial localization of TLSs (25). To minimize false positives, we also carefully checked the TLS scores by cross-referencing them with spot-level expression of T cell marker (*CD3D*) and B cell marker (*MS4A1*) without imputation and the identified TLSs were further carefully reviewed by our experienced pathologists for accuracy.

Unsupervised TLS clustering was then performed based on calculated TLS scores. Prior to clustering, TLS scores were smoothed using a moving average over consecutive windows of 32×32 pixels to reduce noise and highlight local trends in the ST data. The smoothed scores were then clustered using the k-means algorithm, with $k = 5$ selected as the optimal number of clusters based on manual inspection. Clusters containing fewer than 20 pixels were subsequently discarded to enhance robustness and minimize the influence of noise. Following clustering, segmented TLS regions of the corresponding H&E image were carefully reviewed and confirmed by our experienced pathologists to ensure that each segmented area captured a TLS. To segment individual TLSs, we applied connected component labeling using the “label” function from the *scipy.ndimage* module in SciPy (v1.10.1) (95) with default settings in Python. Each TLS was then assigned a unique identifier, along with its corresponding ST data and paired H&E image, facilitating consistent tracking of individual TLSs across samples and downstream analyses. Lastly, TLS size was calculated by measuring the number of pixels within each segmented region. Given that each pixel corresponds to an area of $0.5 \times 0.5 \mu\text{m}^2$, the total area of each TLS was computed by multiplying the pixel count by $0.25 \mu\text{m}^2$.

TLS classification and transcriptomic profiling

To characterize TLSs at different maturation states based on their transcriptomic profiles, we applied log2-transformation followed by z-score normalization across all TLSs on a per-gene basis. This approach stabilized variance and enabled cross-sample comparability while preserving the relative expression dynamics essential for downstream clustering and TLS maturation classification. Following normalization, TLSs were stratified by maturation state based on the spatial expression patterns of key follicular and germinal center (GC) markers. For each TLS, we quantified the expression of follicular dendritic cell (FDC) markers *CR2* and *FCER2* by computing the 95th percentile of their expression values across all pixels within the TLS. These marker values were then used as input for k-means clustering ($k=2$) to classify TLSs into early TLSs (E-TLSs), which lack FDC presence, and mature TLSs (mTLSs), which show strong FDC marker expression. To further resolve the maturation states of mTLSs, we applied a similar strategy using GC-associated markers *MKI67* and *BCL6*, again computing the 95th percentile expression values within each TLS. A second round of k-means clustering ($k=2$) was performed, further classifying mTLSs into P-TLSs, which lack GC-related marker expression (e.g., *BCL6*, *MKI67*), and S-TLSs, which show high GC-related marker expression.

To characterize cellular and signaling changes across TLSs at different maturation states, we analyzed the expression of curated canonical markers and gene signatures corresponding to B cells, T cells, and their phenotypic and functional subsets as well as DCs, macrophages, endothelial cells, and fibroblasts. We also assessed the expression changes in cytokines, chemokines, and genes involved in signaling pathways relevant to TLS formation and maintenance. Marker sets were curated from prior studies (96–100) and refined through manual review to ensure accurate representation of immune and stromal populations. To highlight key differences across TLS maturation states, we selected

26 genes showing significant expression changes among the three maturation states for detailed visualization. Heatmaps were generated using the ‘Heatmap’ function from the ComplexHeatmap R package (101), and boxplots were constructed using the ‘ggplot’ function from the ggplot2 package (102) in R.

Pseudo-Visium spot synthesis from Xenium data

To simulate Visium-style data from Xenium outputs, pseudo-Visium H&E images were first cropped from Xenium H&E images. Pixel coordinates were converted to micrometers using a calibrated scale of 0.85 μm per pixel. A hexagonal spot lattice emulating the geometry of Visium v1/v2 (55 μm spot diameter; 27.5 μm radius) was then tiled across the ROI. Grid centers were generated in image coordinates with half-row staggering, and the grid was initiated half a spot beyond the image boundaries to ensure full peripheral coverage. Only spots whose circular bounding areas intersected the ROI were retained. For each spot, array row and column indices, center coordinates in both pixels and micrometers, spot diameter, and nominal spacing were recorded in a CSV file. Xenium per-molecule transcript coordinates were then assigned to pseudo-Visium spots by mapping each transcript to the nearest spot center within a 27.5 μm radius. Spot-level gene expression matrices were generated by summing transcript counts per gene within each spot. Unless specified otherwise, no molecule weighting or deconvolution was applied beyond this in-radius assignment. Quality control steps removed spots with low total transcript counts (e.g., <200 UMIs), and nontissue areas were optionally excluded using an H&E-derived tissue mask prior to counting. The resulting outputs included: (i) a spot-location file containing pixel and micrometer coordinates for all synthesized spots; (ii) a spot-by-gene count matrix formatted for downstream Visium-style analysis; and (iii) pseudo-Visium H&E images.

Xenium cell type and state identification

Output data from Xenium 5K assays, including gene count matrices, metadata, cellular centroid and boundary coordinates, and transcript coordinates, were processed for transcriptomic and spatial analyses. Initial quality control removed cells with fewer than 30 detected genes as well as cells lacking expression of canonical lineage markers. Data were normalized using the ‘SCTransform’ function in Seurat (v5.1.0) (103). PCA was performed on highly variable genes, and elbow plots were used to determine the number of dimensions for downstream analysis. A nearest-neighbor graph was constructed with ‘FindNeighbors’, followed by Louvain clustering using ‘FindClusters’. UMAP was applied for non-linear dimensional reduction and visualization (103, 104).

Differentially expressed genes (DEGs) for each cluster were identified with ‘FindAllMarkers’, and top DEGs were reviewed alongside canonical lineage markers as reported previously (97, 98, 105, 106). Major cell compartments were annotated, including T cells (e.g., *CD3D*, *CD2*), B and PCs (*MS4A1*, *CD79A* for B cells; *MZB1*, *XBP1* for PCs), myeloid cells (*CD68*, *CD163* for macrophages; *CSF3R* for neutrophils; *KIT*, *MS4A2* for mast cells; *CD1A*, *CD1C*, *LAMP3* for DCs), stromal cells (*COL5A1*, *PDGFRA* for cancer-associated fibroblasts (CAFs); *VWF*, *PECAM1*, *CD34* for endothelial cells; *MYH11* for smooth muscle cells), epithelial cells (*EPCAM*, *CDH1*), and tumor cells (*CDK1* and *MKI67*). In addition, regulatory B cells (Bregs) were further identified based on *IL10* gene expression (107). Cell type annotations were mapped back to tissue images in Xenium Explorer for visualization and validation against cellular morphology.

Xenium TLS identification

TLS identification in Xenium samples followed a two-stage strategy depending on the availability of an adjacent Visium ST section. When an adjacent Visium ST section was available, TLSs were first identified using our ST-based TLS detection framework and manually verified on the corresponding Xenium slide. Otherwise, TLSs were delineated

on the H&E image using the H&E-based TLS detection algorithm described below. The H&E section was then co-registered to the Xenium morphology image in Xenium Explorer. We exported the affine transformation matrix from Xenium Explorer and used it to map TLS segmentation coordinates from the H&E image space into the Xenium single-cell coordinate space. Formally, for a point in H&E pixel coordinates, the transformed Xenium coordinates were computed using a homogeneous-coordinate affine transformation:

$$\begin{bmatrix} x_{\text{Xenium}} \\ y_{\text{Xenium}} \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & t_x \\ a_{21} & a_{22} & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{\text{H\&E}} \\ y_{\text{H\&E}} \\ 1 \end{bmatrix}$$

TLS labels were then assigned at single-cell resolution by determining whether each Xenium cell boundary, or alternatively its centroid, fell within the transformed TLS polygons.

Tumor spot identification

To identify tumor spots from Visium ST data, we applied SpatialInferCNV (108) (<https://aerickso.github.io/SpatialInferCNV/>) in R, based on the inferCNV framework (108), as we previously described (75). The analysis was performed using the following parameters: cutoff = 0.1, cluster_by_groups = FALSE, HMM = FALSE, and denoise = TRUE. Specifically, we followed a multi-step approach as outlined in our recent study (75). To ensure reliable results, we employed an integrative approach involving multiple rounds of selection to identify suitable non-tumor reference spots for copy number variant (CNV) calling. Reference spots were selected based on two criteria: (i) histological annotation as “non-tumor” and (ii) transcriptional distinctiveness from tumor spots, as indicated by UMAP embedding (i.e., topographically distant from tumor cell clusters). To ensure that we selected true “non-tumor” spots without contamination of tumor cells, we further excluded any reference spots that displayed evidence of CNV events despite being histologically labeled as “non-tumor.” For CNV benchmarking, we selected at least two independent reference populations and performed Pearson correlation coefficient analysis to evaluate their concordance at the chromosomal band level across subclones. To ensure data quality, samples with weak or ambiguous inferCNV signals were excluded, and only those with reliable CNV profiles were retained for downstream analysis. For samples lacking CNV signals or identified as chromosomally stable tumors, tumor regions were manually annotated by our experienced pathologists using the corresponding H&E images.

TLS location identification

To categorize the spatial relationship between TLSs and tumor areas, we overlaid TLS coordinates with tumor spot annotations. TLSs located entirely within tumor spots were annotated as IT TLSs. TLSs positioned outside tumor spots but within 200 μm of the tumor-stromal interface (defined by the nearest tumor spots) were labeled as PT TLSs. TLSs situated beyond this proximity threshold were classified as DT TLSs. TLS location categories were further confirmed by our experienced pathologists. For samples with no or weak inferred CNV signals (i.e., chromosomally stable tumors), the spatial relationship between TLSs and tumor regions was determined based on pathology annotations of the corresponding H&E images. Pathology annotations for BRCA samples were obtained from the original study (57).

Quantifying distance from an IT or DT TLS to surrounding tissue spots

To quantify the spatial relationship between TLSs and their surrounding microenvironment, TLS pixel coordinates were first converted to ST spots. Radial distances from the center of each TLS to all the spots were then calculated using the ‘RadialDistance’ function from the Semla (v1.2.1) package (109). For intra-tumoral TLSs (IT TLSs), only tumor spots were retained, while for distal-tumoral TLSs (DT TLSs) located in histologically normal regions, only non-tumor spots were

retained. In both cases, distance values were rescaled to a 0–1 range to enable comparative distance analysis across samples. This standardized distance metric enabled us to investigate TLS proximity to tumor regions and assess potential associations between TLS location and TME characteristics.

Pathway analysis

To investigate tumor-TLS interactions, we quantified the activity of tumor-associated pathways in tumor-enriched spots using curated gene sets. These included Hallmark gene sets from the Molecular Signature Database (MSigDB), which represent key biological processes such as cell cycle, oncogenic pathways, metabolic related pathways, and antigen presentation (110). We also calculated scores for tumor meta-programs that define key transcriptional states in tumors, including cell cycle regulation, stress response, EMT, antigen presentation, oncogenic signaling, metabolism, and lineage-specific differentiation (111). Additionally, we evaluated gene modules representing recurrent cancer cell states across tumor types, such as interferon response, hypoxia, oxidative phosphorylation, metal response, and EMT (77). Signature scores were computed using the 'AUCell_run' function from the AUCell package (112), which ranks genes by expression within each spot and calculates the Area Under the Curve (AUC) for recovery of each gene set within the top-ranked genes. Scores were calculated separately for each sample using log-normalized expression matrices, without additional scaling or centering, as AUCell relies on gene rankings rather than raw expression values.

To investigate spatial changes in pathway activity, we modeled pathway scores in relation to TLS proximity using the 'differentialGeneTest' function from the Monocle (v2.34.0) package in R (113). Using the function's default settings, we employed a natural spline with three degrees of freedom to fit pathway scores as a function of spatial distance from TLS. This approach allowed us to identify pathways exhibiting significant expression variation across TLS proximity gradients.

For pathways identified as significantly associated with TLS proximity, we generated smoothed expression curves using the 'geneSmoothCurves' function from the Monocle package to discern expression trends with respect to TLS proximity. Similar to the 'differentialGeneTest' function, we employed a natural spline with three degrees of freedom. Pathway scores were smoothed across spatial distance from IT TLSs and z-score normalized for each pathway. Scores from spots nearest to the TLS (distance = 0), and the farthest spots (distance = 1), were used to classify expression trends. A score was considered low if the z-score was less than or equal to zero and high if it was greater than or equal to zero, relative to the pathway's mean across all tumor-enriched spots distributed along the entire spatial gradient from the TLS. Based on the resulting profiles, we identified two major trends: high-to-low, where expression was elevated near a TLS and declined with distance, and low-to-high, where expression increased with distance from a TLS.

To identify spatially enriched pathways that were consistently observed across cancer types, we first aggregated samples by cancer type. We then identified pathways showing significant high-to-low or low-to-high trends within each type based on our differential analysis. For each pathway, we computed an enrichment proportion, defined as the fraction of cancer types in which the pathway showed significant spatial enrichment. We selected the five pathways with the highest enrichment proportions from each trend category (high-to-low and low-to-high), yielding the top 10 pathways with the most consistent spatial expression patterns in relation to TLS proximity. To visualize these patterns, we generated heatmaps using the 'plot_pseudotime_heatmap' function in Monocle (v2.34.0) (113) and overlaid line plots using the 'ggplot' function in ggplot2 in R to display the smoothed pathway scores along the IT TLS distance gradient (102).

Identifying ligands and receptors enriched within IT TLSs

To quantify cytokines and chemokines enriched inside intra-tumoral TLSs (IT TLSs) relative to their immediate tumor margins, we first computed

spot-wise radial distances to each TLS using the 'RadialDistance' function in Semla (v1.2.1) (109). Spots with distance < 0 were classified as TLS core (inside), while the TLS margin was defined as spots located within 0–50 μm from the TLS boundary. When no spots were present in this window, a fallback margin of 0–100 μm was applied, and all analyses were restricted to tumor-annotated spots. For each intra-tumoral TLS, we compared gene expression between TLS core and TLS margin spots across cytokine and chemokine ligands and receptors. For every gene, we calculated the mean expression, the proportion of expressing spots, and the difference in mean expression (core – margin). Paired Wilcoxon signed-rank tests were then performed across TLSs to assess whether expression was significantly higher inside TLSs compared with their margins.

Gene-level analysis of spatial gradients relative to TLS proximity

To investigate spatial expression gradients of individual cytokine and chemokine ligands and receptors with respect to intra-tumoral TLSs, we applied the same procedure used for modeling pathway scores in relation to TLS proximity. Spot-wise distances from TLSs were first computed using the 'RadialDistance' function in Semla (v1.2.1) (109). Gene expression trends were then modeled with the 'geneSmoothCurves' function in Monocle (v2.34.0) using a natural spline with three degrees of freedom, followed by z-score normalization per gene. Smoothed values across the TLS-tumor distance gradient were used to classify spatial profiles: expression that declined from TLS core (distance = 0) to distal tumor regions (distance = 0.75) was labeled “high-to-low” (113). To identify robust signals, results were aggregated across eight cancer types, and genes showing concordant trends in multiple types were prioritized.

Tissue annotation, TLS spatial distribution, and tissue enrichment analysis

Spot-level pathologist annotations for tumor samples in the CRC cohort were obtained from the original study (38). For the BLCA cohort, spot-level histology annotation was performed by our board-certified pathologists. For the LIHC cohort, we examined the region-specific distribution of immune cells, stromal cells, and tumor cells by categorizing all ST spots into three groups based on their proximity to the tumor region: (i) Tumor region spots, which are entirely or predominantly composed of neoplastic cells; (ii) tumor boundary spots, located within the stroma immediately adjacent to the tumor region (typically the first one or two layers of spots, approximately within 200 μm of the tumor-normal interface); and (iii) nontumor spots, which are topographically distal from the tumor region and do not fall into either the tumor region or tumor boundary category, as recently described (75). To determine the spatial distribution of TLSs within tissue compartments annotated by pathologists, we overlaid TLS coordinates onto the pathologist-defined annotations. This allowed us to systematically determine the tissue compartment in which each TLS was located.

To evaluate the enrichment or depletion of TLSs in specific tissue compartments, we applied the $R_{o/e}$ (ratio of observed to expected) approach, as previously described (98, 114). In short, we computed the ratio of the observed TLS count to its random expectation, using a χ^2 test for each TLS maturation state across different tissue compartments. An $R_{o/e}$ value greater than 1 indicates enrichment, whereas an $R_{o/e}$ value less than 1 suggests depletion of TLSs in a given tissue compartment.

Sample classification and correlative analysis with TLS features

To investigate associations between TLS maturation and clinical or molecular characteristics, we first quantified TLS features, including total TLS count and the relative abundance of each maturation state. Each sample was then classified into one of three categories, E-TLS, P-TLS, or S-TLS, based on which maturation state was most prevalent within the sample. Clinical variables evaluated across cohorts included tumor grade, tumor stage, KRAS and MTAP mutation status, HBV

infection status, presence of metastasis, and chemotherapy response. Associations involving total TLS count and relative abundance were assessed using the Chi-square test, whereas relationships between TLS maturation-based sample classification and clinical features were evaluated using proportion tests.

WSI analysis, TLS detection, and TLS maturation classification

LUSC, KIRC, BLCA, LUAD, STAD, and COAD WSIs were obtained from the Genomic Data Commons (GDC) portal ([115](https://portal.gdc.cancer.gov/)) (<https://portal.gdc.cancer.gov/>) and processed to identify and classify TLS maturation states. TLS detection was performed using HookNet-TLS ([47](#)), a deep learning-based segmentation model designed to detect TLSs in histopathological images. The model takes as input WSIs at two resolutions, high (0.5 $\mu\text{m}/\text{pixel}$) and low (2.0 $\mu\text{m}/\text{pixel}$), along with binary tissue masks indicating valid tissue areas. Default parameters were used, including a patch size of 512×512 pixels, a stride of 256 pixels, and class-specific probability thresholds optimized via grid search to maximize F1-score. Although the model predicts multiple tissue classes, only TLS-predicted regions were retained for downstream analysis. To distinguish individual TLSs, clusters of connected TLS-positive pixels were extracted from the binary prediction mask using the 'findContours' function from the OpenCV package in Python. Each region was assigned a unique identifier based on its order of detection, and the corresponding spatial coordinates were stored for downstream analysis. In total, 21,955 TLSs were identified across all samples (LUSC, $n=6,688$; BLCA, $n=2,538$; KIRC, $n=899$; LUAD, $n=6,049$; STAD, $n=3,484$; COAD, $n=2,297$). To ensure accurate visualization and morphological assessment, spatial coordinates were mapped to WSIs and manually reviewed. For each TLS, the coordinates were retrieved and rasterized into a binary mask using the 'fillPoly' function from the OpenCV package. This mask was then used to extract the corresponding area from the whole-slide image, and the bounding rectangle was computed using the 'boundingRect' function in OpenCV to segment the TLSs. As with TLSs identified in ST data, TLS size in WSIs was calculated by determining the number of pixels in each segmented H&E image, where each pixel corresponds to an area of $0.25 \times 0.25 \mu\text{m}^2$. The total area of each TLS was then computed by multiplying the pixel count by $0.0625 \mu\text{m}^2$.

Following TLS detection, a two-step YOLOv8-based classification pipeline ([48](#)) was developed to categorize TLSs by maturation states. The model was trained on a subset of data from three TCGA cohorts (LUSC, BLCA, and KIRC), comprising 956 TLS images. To reduce batch effects due to differences in staining, slide scanning, and tissue processing, YOLOv8 was trained using the default data augmentation parameters, including flips, rotations, scaling, cutout, mix-up, brightness and contrast shifts, hue and saturation jitter, and Gaussian blur and noise. In the first step, TLSs were classified as E-TLS or mTLS. In the second step, mTLSs were further sub-classified into P-TLSs and S-TLSs. Two separate classifiers were trained, one for E-TLS vs. mTLS and another for P-TLS vs. S-TLS. For each cancer type, TLS images were manually annotated, stratified by cancer type, and randomly split into 80% training and 20% test sets. Models were trained using image-based features extracted from the segmented TLSs. Both classifiers were trained for 200 epochs using stochastic gradient descent with momentum and cosine learning-rate scheduling. Performance was evaluated on the held-out 20% test sets using accuracy, precision, recall, and F1-score, with training and test loss and accuracy monitored across epochs to confirm stable convergence. Specifically, precision measured the proportion of correctly predicted positive instances out of all predicted positives, recall measured the proportion of correctly predicted positives out of all actual positives, and F1-score represented the harmonic mean of precision and recall. The numbers of training and testing images for each cancer type, along with a summary of classification metrics, are provided in fig. S5, A, B, and D.

To directly evaluate the cross-cancer generalizability of the YOLOv8-based TLS maturation classifier, we applied the trained YOLO model to matched H&E images from our combined ST cohorts. Importantly, none of these H&E images were included in model training and were therefore entirely unseen by YOLO. Model predictions were evaluated against ST-based TLS annotations. No retraining, fine-tuning, or threshold recalibration was performed; the model was applied as-is using the same preprocessing steps, parameter values, and operating thresholds learned during training, ensuring a stringent and unbiased assessment of generalizability. Following the same two-step workflow used throughout the manuscript, E-TLSs were first distinguished from mature TLSs (mTLSs), after which mTLSs were further subclassified into P-TLSs and S-TLSs. Model performance was quantified using accuracy, precision, recall, and F1 score, with receiver operating characteristic (ROC) and precision-recall analyses used to assess discriminative performance under class imbalance.

Therapy-associated WSI cohorts

To characterize TLS maturation in therapy-associated settings, four independent clinical cohorts spanning immunotherapy, targeted therapy, chemotherapy, and combined chemo-targeted regimens were analyzed ([43–46](#)). An independent in-house OCCC cohort included 34 platinum-resistant patients treated with combined anti-CTLA-4 and anti-PD-1/PD-L1 immunotherapy, with high-resolution H&E images available for 26 patients (25 pretreatment and 14 on-treatment samples) ([43](#)). Two breast cancer cohorts consisted of diagnostic biopsy H&E slides collected before neoadjuvant systemic therapy. The HER2-targeted therapy cohort included 85 HER2⁺ invasive breast carcinoma cases confirmed by ASCO/CAP IHC/ISH criteria, with treatment response defined as pathological complete response versus residual invasive disease. Patients received neoadjuvant trastuzumab $\pm$ pertuzumab; 37 patients were responders and 48 patients were non-responders ([46](#)). The neoadjuvant chemotherapy cohort comprised 197 patients prospectively enrolled into the TransNEO study and treated with block-sequential taxane plus anthracycline for a median of 18 weeks, with response evaluated using the Residual Cancer Burden (RCB) system, yielding 57 responders and 140 non-responders ([45](#)). The ovarian cancer cohort consisted of post-treatment surgical resection H&E slides from 78 high-grade serous ovarian cancer (HGSOC) patients treated with carboplatin/paclitaxel plus bevacizumab. Response was assessed using post-treatment CA-125 kinetics and measurable tumor regression, identifying 26 responders and 52 non-responders ([44](#)). All H&E images were processed using the same pipeline described above. Across these cohorts, we identified 3,133 TLSs in total [OCCC, $n=91$ ([43](#)); HGSOC, $n=2,236$ ([44](#)); BRCA, $n=503$ ([45](#)); BRCA, $n=303$ ([46](#))]. By maturation state, this included 1,346 E-TLSs [OCCC, $n=70$ ([43](#)); HGSOC ([44](#)), $n=799$; BRCA, $n=267$ ([45](#)); BRCA, $n=210$ ([46](#))], 1,627 P-TLSs [OCCC, $n=18$ ([43](#)); HGSOC ([44](#)), $n=1,326$; BRCA, $n=201$ ([45](#)); BRCA, $n=82$ ([46](#))], and 160 S-TLSs [OCCC, $n=3$ ([43](#)); HGSOC ([44](#)), $n=III$; BRCA, $n=35$ ([45](#)); BRCA, $n=11$ ([46](#))].

TLS group stratification

To assess the prognostic value of TLS maturation states, we computed a composite TLS score that integrates the relative contributions of each TLS state, E-TLS, P-TLS, and S-TLS, for each sample. Absolute counts of each TLS state were first quantified per sample, resulting in a distribution of TLS maturation states. To mitigate skewness and address overdispersion in the raw counts, we applied a log1p transformation to each state count. Samples lacking TLSs were assigned a composite score of zero to reflect the absence of TLS formation. For TLS⁺ samples, we performed PCA on z-score normalized, log-transformed TLS state counts within each cancer cohort to identify dominant axes of variation. Each TLS⁺ sample was then projected into this PCA space, and a composite score was calculated as a weighted sum of the first two principal components (PC1 and PC2), with weights proportional to the

variance explained by each component. This data-driven approach captures both the abundance and compositional diversity of TLS maturation in an orthogonal and continuous manner. For downstream survival analysis, all samples were stratified into C1 and C2 groups based on the median value of the composite TLS score.

Survival analyses

Survival analysis was performed to assess the association between TLS maturation states and OS across different cancer types. KM survival curves were generated using the 'Surv' and 'survfit' functions from the survival package (116) in R, and statistical differences between groups were assessed via the log-rank test. For TCGA cohorts, clinical annotations and OS times were obtained from the PanCanAtlas study (117). To estimate the impact of TLS maturation on survival while adjusting for potential confounders, including tumor stage, multivariable Cox proportional hazards models were fit using the 'coxph' function from the survival package (116), from which hazard ratios (HRs) and 95% confidence intervals (CIs) were computed.

Additional statistical analyses

In addition to the algorithms described above, all other basic statistical analyses were performed in the R statistical environment (v4.3.0, v4.4.1). Enrichment across various conditions was assessed using a chi-square test on observed counts, followed by calculation of standardized and adjusted residuals to quantify deviations from expected values. Adjusted residuals were used to highlight significant patterns, with corresponding *P*-values indicating statistically significant enrichment or depletion across different conditions within the dataset. Pairwise comparisons of gene expression were conducted using the Wilcoxon rank-sum test to assess statistical significance between TLS maturation states. Spearman's correlation coefficient was used to evaluate the association between two continuous variables, such as distance from TLSs and pathway scores in tumor spots. The correlation matrix was generated using the 'cor' function with the 'spearman' method in R. Proportion tests were conducted separately for each TLS maturation state to examine its relationship with clinical features. All statistical tests in this study were two-sided. To account for multiple hypothesis testing, *P*-values were adjusted using the Benjamini-Hochberg method to calculate false discovery rate (FDR) *q*-values. Results were considered statistically significant at a *P*-value or FDR *q*-value threshold of <0.05.

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Atlas Network (HTAN) data portal (<https://humantumoratlas.org>). Details on data and code availability, along with metadata associated with each cohort, are provided in table S1, table S2, and the materials and methods section. No new reagents were generated for this study. The computational framework is described in the materials and methods, and the analysis code is available in the GitHub repository at https://github.com/Coolgenome/Pan-Cancer_Spatial_Atlas_TLS. Any additional information required to reanalyze the data reported in this paper is available from the corresponding author upon request. **License information:** Copyright © 2026 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.science.org/about/science-licenses-journal-article-reuse>

SUPPLEMENTARY MATERIALS

[science.org/doi/10.1126/science.adz2742](https://doi.org/10.1126/science.adz2742)
Figs. S1 to S8; Tables S1 to S8; Reproducibility Checklist
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Pan-cancer spatial atlas of tertiary lymphoid structures

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Editor's summary

Tertiary lymphoid structures (TLSs) form local immune hubs inside tumors, but they are diverse and not all are equally functional. Cho *et al.* built a pan-cancer atlas and developed an artificial intelligence (AI)-based framework to detect and characterize TLSs in human tumors. TLSs varied in maturation state, spatial location, cellular composition and organization. Intratumoral TLSs were associated with spatial gradients in tumor-intrinsic signaling. A scalable model was trained to detect and classify TLSs on standard pathology slides, and a composition-based TLS score was designed to stratify patients based upon survival outcomes. The study provides insights into TLS biology and may offer a path toward integrating TLS features into future clinical trials. —Priscilla N. Kelly

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